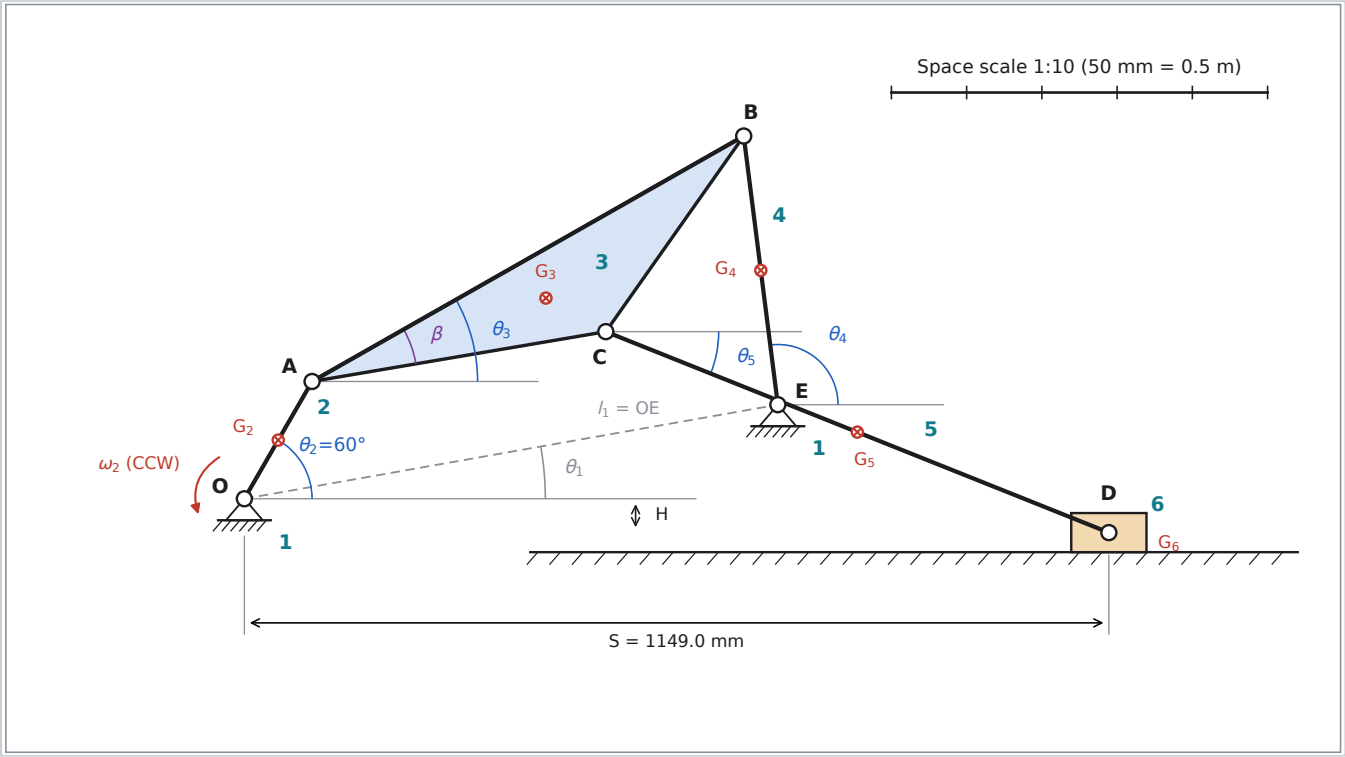
Kinematic and Dynamic Analysis of a Six-Bar Linkage Mechanism

Major Assignment — fully worked practice solution



Prepared for	Learning / exam preparation (not for submission)
Student	_____ Student no. _____
Parameters	OA = 180 mm, β = 20°, θ_1 = 10°, n_2 = 440 rpm (example values — replace with the ones your lecturer gives you)
Methods	Graphical · Instantaneous centres · Analytical (complex-number vector loops) · Computational (multibody simulation) · Matrix force analysis · Balancing
Date	October 2026

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Summary

This report analyses the six-bar linkage of the MIET1077 major assignment: a crank (link 2) drives a ternary coupler (link 3) that is held by a rocker (link 4); a connecting rod (link 5) taken from point C of the coupler drives a slider (link 6) on a horizontal guide. The crank length is OA = 180 mm and it turns anticlockwise at a constant 440 rpm (ω_2 = 46.077 rad/s). All other lengths follow from the course data-selection rules (OE = CD = 4·OA, AC = 3/5·AB, H = OA/4) with AB = 660 mm and BE = 360 mm chosen to give a Grashof crank–rocker with a good transmission angle (never below 54.8°).

What the mechanism does. The crank turns fully, the rocker swings through 60.0°, and the slider makes a 299.1 mm stroke between S = 924.5 mm and 1,223.6 mm from the crank pivot. The slider's rightward stroke takes 193.6° of crank rotation and its return takes 166.4°, a quick-return time ratio of 1.16.

Key results at the specified position (θ_2 = 60°).

Quantity	Value
Slider velocity / acceleration	7.01 m/s ← and 181.2 m/s² ← (both to the left)
Angular velocities ω_3, ω_4, ω_5	8.22 (CW), 12.60 (CCW), 1.40 (CW) rad/s
Angular accelerations α_3, α_4, α_5	432.8, 1,059.2, 248.2 rad/s² (all CCW)
Driving torque on the crank T_{12}	166.0 N·m
Largest pin force (crank bearing O)	1,590 N
Slider side load on the guide	215 N
Shaking force / shaking moment on the frame	1,813 N at 29.9° / 85.4 N·m

Over one revolution the driving torque swings between −184 and +185 N·m (mean zero, because there is no load or friction), the crank bearing sees up to 2.92 kN near $\theta_2 \approx 20°$, and the shaking force reaches 3.07 kN (RMS 1.80 kN).

Agreement between methods. The graphical polygons, read with ruler precision (±0.5 mm), agree with the analytical vector-loop solution to within 0.8 % for velocities and 0.2 % for accelerations. The instantaneous-centre method agrees to within 1.4 %; its largest error comes from reading a short (≈15 mm) radius at 1:10 scale. An independent multibody simulation, built the way Working Model builds a model (bodies, pins, slot, velocity motor) and integrated in time, reproduces every analytical quantity, including all pin forces and the motor torque, to better than 10^{-7} over the whole revolution.

Balancing recommendation. Counterweights on the two ground-pivoted links reduce the RMS shaking force by 77 % with the exact Berkof–Lowen (four-bar) solution, and by 87 % with a least-squares optimum. The residual is the reciprocating inertia of the slider plus half of link 5, which no rotating counterweight can cancel completely. The cost is extra moving mass (about 10.9 kg at 120 mm radius) and a larger torque fluctuation, so a flywheel should be specified together with the counterweights.

Objectives

- Select the dimensions of the linkage from the given data using the course data-selection (non-equality) equations and check mobility, the Grashof condition and the transmission angle.
- Determine all positions of the links and joints graphically for 16 crank positions, the joint paths and the limits of motion.
- Determine linear and angular velocities and accelerations at $\theta_2 = 60°$ graphically (vector polygons drawn to scale).
- Locate all 15 instantaneous centres with Kennedy's theorem and use them to find the velocities of A, B, C and D.
- Derive and solve the analytical (vector loop, complex number) equations for position, velocity and acceleration of all five moving links, and compare them with the graphical results.
- Set up and solve the 14×14 matrix of dynamic joint forces and the driving torque at $\theta_2 = 60°$, from free-body diagrams of every link.
- Build a computational model of the mechanism and use it to compute paths, velocities, accelerations, pin forces, slider side load and driving torque over one revolution, and compare it with the other methods.
- Determine the shaking force and shaking moment and develop an analytical strategy for balancing the mechanism.

Related Theory

Mechanism, mobility and Grashof's criterion. A mechanism is an assembly of rigid links joined by kinematic pairs that transmits a constrained motion (Week 1; Subic & Alam 2010). The degree of freedom of a planar mechanism follows from the Gruebler–Kutzbach equation,

$$\mathrm{DOF} = 3(n-1) - 2P_1 - P_2 = 3(6-1) - 2(7) - 0 = 1,$$

with n = 6 links (including the frame), P_1 = 7 full joints (pins O, A, B, C, D, E and the slider pair) and P_2 = 0 half joints. One input, the crank angle θ_2, therefore fixes the configuration. Grashof's criterion, s + l ≤ p + q, guarantees that the shortest link of a four-bar can rotate fully (Week 1). When the shortest link is the crank next to the frame, the four-bar O-A-B-E is a crank–rocker.

Types of motion. Link 2 rotates about a fixed axis, link 4 oscillates about a fixed axis, links 3 and 5 have general plane motion, and the slider (link 6) translates rectilinearly (Supplementary Guide, Figure 1).

Relative-motion (graphical) method. For two points A and B of the same rigid link,

$$\mathbf{V}_B = \mathbf{V}_A + \mathbf{V}_{B/A}, \qquad |\mathbf{V}_{B/A}| = \omega\, r_{AB}, \quad \mathbf{V}_{B/A} \perp AB,$$

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}^{n}_{B/A} + \mathbf{a}^{t}_{B/A},$$

$$a^{n}_{B/A} = \omega^2 r_{AB} = \frac{V^2_{B/A}}{r_{AB}}\ (\text{directed from B towards A}), \qquad a^{t}_{B/A} = \alpha\, r_{AB}\ (\perp AB).$$

Absolute vectors start from the pole of the polygon. Vectors of points on ground-connected links start from the pole. Every normal component must be calculated before an acceleration polygon can be closed (Week 3). The polygon points of one link form an *image* similar to the link, which gives the velocity and acceleration of any further point, such as a mass centre.

Instantaneous centres. An instantaneous centre (IC) is the point common to two links at which they have no relative velocity. A mechanism of n links has $C = n(n-1)/2$ ICs. Pins are permanent ICs, and the IC of a slider and its straight guide lies at infinity perpendicular to the guide. The *Aronhold–Kennedy theorem* states that the three ICs of any three links lie on one straight line, which locates the remaining ICs (Week 5). The velocity of any point is then $V = \omega\, r_{IC}$, perpendicular to the radius from the IC.

Analytical (complex-number vector-loop) method. Every link is a vector $\mathbf{R} = l\,e^{j\theta} = l(\cos\theta + j\sin\theta)$ (Euler's identity). Writing each closed loop as a vector sum equal to zero and separating real (x) and imaginary (y) parts gives the position equations. Differentiating once, with $\frac{d}{dt}e^{j\theta} = j\omega e^{j\theta}$, gives the velocities, and differentiating again gives the accelerations, with $\frac{d}{dt}(j\omega e^{j\theta}) = (j\alpha - \omega^2)e^{j\theta}$ (Week 7).

Dynamic force analysis (matrix method). Newton–Euler equations for each rigid link, $\sum \mathbf{F} = m\mathbf{a}_G$ and $\sum M_G = I_G\alpha$ (or D'Alembert's form $\sum \mathbf{F} - m\mathbf{a}_G = 0$), written for every moving link, give a linear system $[A]\{B\} = \{C\}$. Here [A] holds the geometry (unit coefficients and the R-vector components), {B} the unknown joint forces and the driving torque, and {C} the inertia terms (Week 9).

Shaking force, shaking moment and balancing. The shaking force is the vector sum of all forces the moving links apply to the frame, $\mathbf{F}_s = \mathbf{F}_{21} + \mathbf{F}_{41} + \mathbf{F}_{61} = -\sum m_i \mathbf{a}_{Gi}$. The shaking moment is the reaction torque plus the moments of these ground forces (Week 10). A linkage is completely force-balanced when the total centre of mass of its moving links is stationary. Berkof & Lowen (1969) showed how to achieve this for a four-bar with counterweights on the two ground-pivoted links. Rotating-mass balancing uses $\sum mr = 0$ (static) and $\sum mra = 0$ (dynamic) (Week 10). Reciprocating masses can only be partially balanced by rotating counterweights.

Computational multibody method. Programs such as Working Model 2D describe each body by the coordinates (x, y, φ) of its mass centre, the joints by algebraic constraint equations Φ(q, t) = 0 and a motor by a driving constraint. They

then solve $M\ddot{q} + \Phi_q^T \lambda = Q$ at every time step. The Lagrange multipliers λ are the joint reaction forces and the motor torque (Uicker et al. 2017; Norton 2012).

Data Selection and Parameters

Parameters given by the lecturer (blank in the handout; each student receives their own). Example values used in this worked solution:

Given	Value
Crank length l_2 = OA	180 mm
Angle of the ternary link β = ∠BAC	20° (C below AB)
Angle of the frame line OE, θ_1	10° above horizontal
Crank speed n_2	440 rpm, anticlockwise, constant ($\alpha_2 = 0$)

Parameters selected by the student with the data-selection (non-equality) equations of Week 1:

$$\overline{OE} = \overline{CD} = 4\,\overline{OA}, \qquad \overline{AC} = \tfrac{3}{5}\,\overline{AB}, \qquad H = \tfrac{\overline{OA}}{4},$$

$$\overline{OA} + \overline{AB} < \overline{OE} + \overline{BE}, \qquad \overline{OA} + \overline{OE} < \overline{AB} + \overline{BE}.$$

Parameter	Equation	Value
l_1 = OE	4·OA	720 mm
l_5 = CD	4·OA	720 mm
H	OA/4	45.0 mm
l_3 = AB	selected	660 mm
l_4 = BE	selected	360 mm
AC	3/5·AB	396 mm
ω_2	$2\pi n_2/60$	46.0767 rad/s

Table 1 – Derived and selected dimensions.

Check	LHS (mm)	RHS (mm)	Satisfied
OA + AB < OE + BE	840	1,080	yes
OA + OE < AB + BE	900	1,020	yes
Grashof: s + l <= p + q	900	1,020	yes
OA is the shortest link (crank-rocker)	180	180	yes

Table 2 – Verification of the selection inequalities and Grashof's criterion.

Why AB = 660 mm and BE = 360 mm? Many pairs satisfy the two inequalities. This pair was chosen because it (i) keeps a comfortable margin on both inequalities, (ii) makes OA the shortest link, so the crank rotates fully (crank–rocker), (iii) keeps the four-bar transmission angle between 54.8° and 89.9° (good practice is > 40°), and the slider transmission angle above 68.0°, and (iv) reproduces the configuration of the handout figure, with BE almost vertical (θ_4 = 97.2°) at θ_2 = 60°.

Mass properties (handout item 6).

Link	Mass (kg)	I_G (kg·m²)	Mass-centre location
2 – crank	1.0	0.002	G_2 at OA/2

Link	Mass (kg)	I_G (kg·m²)	Mass-centre location
3 – ternary link	2.5	0.008	G_3 at AB/2 from A along the bisector of β (β/2 below AB)
4 – rocker	1.5	0.005	G_4 at EB/2
5 – connecting rod	1.8	0.006	G_5 at CD/2
6 – slider	0.9	– (translates)	G_6 at D

Table 3 – Mass properties (handout item 6). G_3 is placed literally as specified, "at β/2 and AB/2": on the bisector of β, AB/2 = 330 mm from A. The Week 9 hand sketch instead marks G_3 at the centroid of triangle ABC, which is also Working Model's default for a polygon. Section 6 shows this choice changes the joint forces by only 1–3 %, but confirm with your lecturer which one is intended.

Coordinate system and sign convention. Origin at the crank pivot O, x horizontal to the right, y vertically up. All angles are measured anticlockwise (CCW) from +x. Positive ω and α are CCW. θ_5 is the angle of the vector C→D. The slider path is the horizontal line y = −H.

1. Graphical Position Analysis

Procedure (Week 1 “Steps for position drawing”). For each crank angle θ_2:

a. With centre O, draw OA at θ_2.

b. With centre O, draw OE at θ_1 (fixed) and mark E.

c. With centre A, strike an arc of radius AB. With centre E, strike an arc of radius BE. Their intersection (the one on the same side as in the handout figure) is B.

d. With centre A, draw AC at angle β below AB and mark C (AC = 396 mm). Join C to B; triangle ABC is link 3.

e. Draw the slider path, a horizontal line H = 45 mm below O.

f. With centre C, strike an arc of radius CD. Its intersection with the slider path (to the right of C) is D.

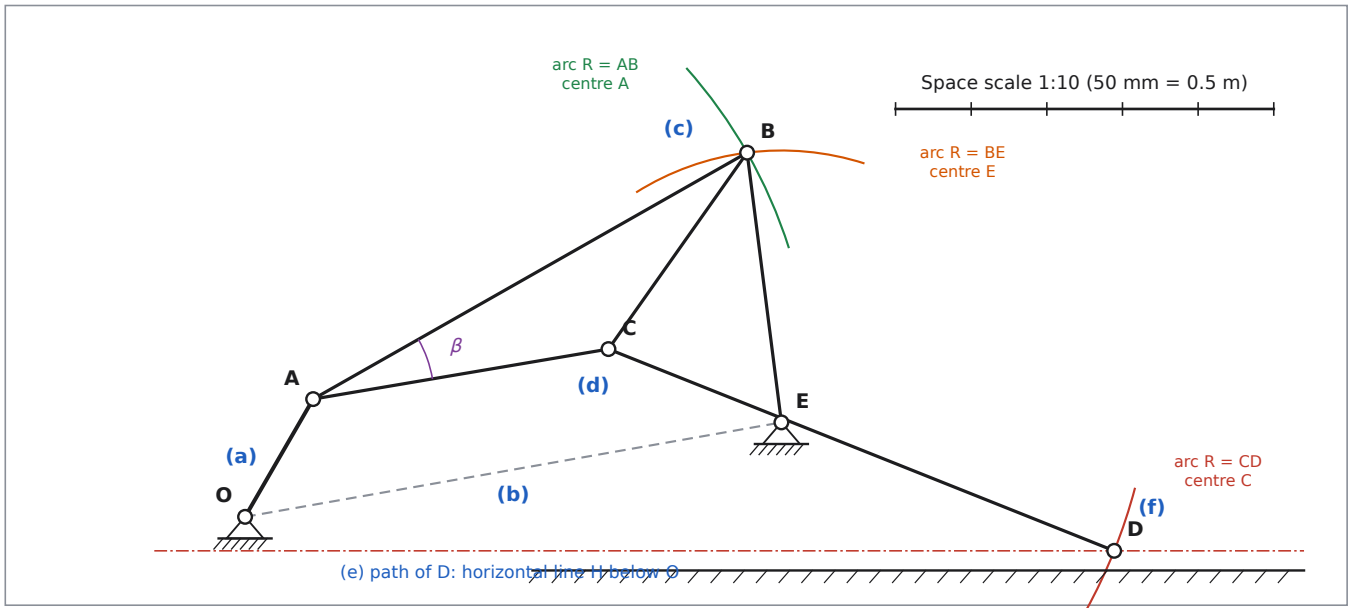


Figure 1 – Sample working for one position (θ_2 = 60°): construction steps (a)–(f). Drawn to scale 1:10 (print at 100 %).

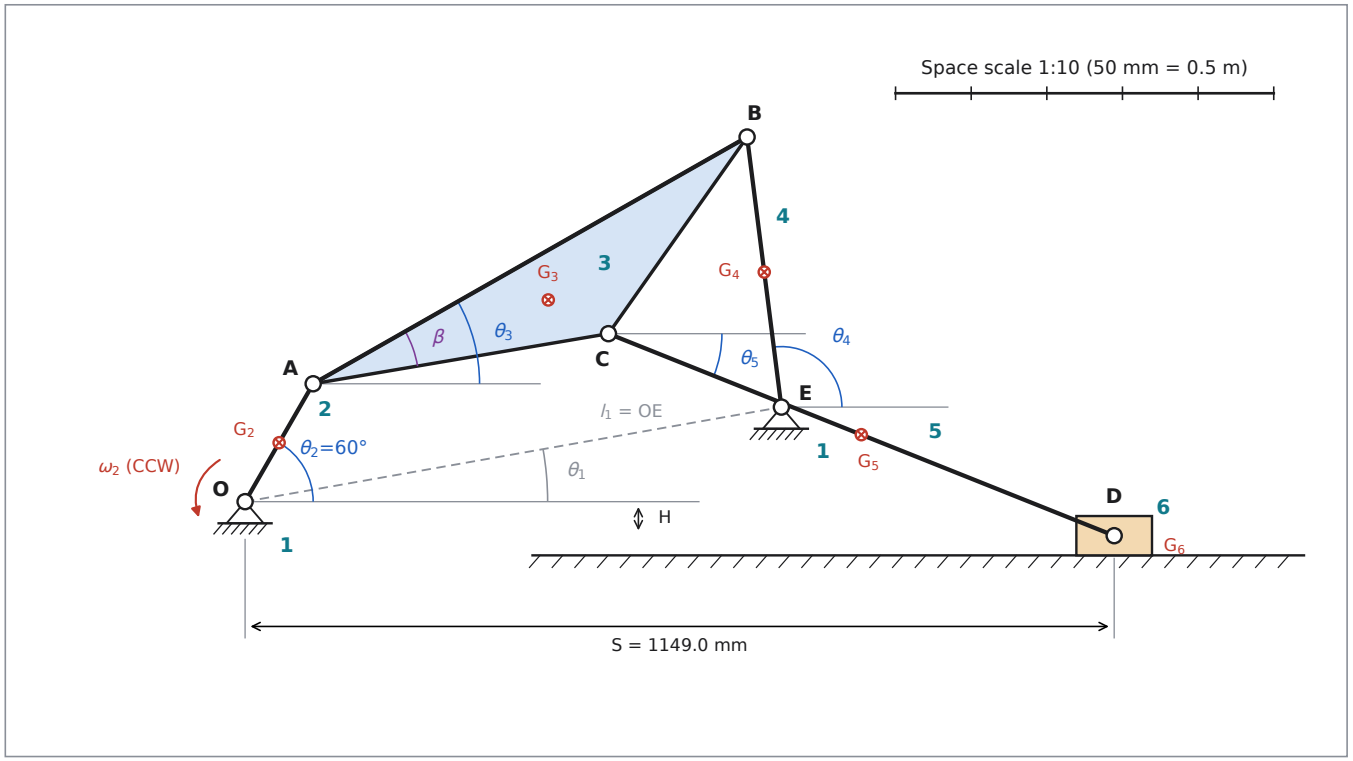


Figure 2 – Space diagram at the given position θ_2 = 60° with link numbers, angle conventions and mass centres G_2–G_6 (scale 1:10).

Repeating the construction every 22.5° gives the 16 subsequent positions of link 2 (position 1 = the given position θ_2 = 60°; positions numbered anticlockwise). The joints of each position are joined to show the path of every moving joint (Figure 3).

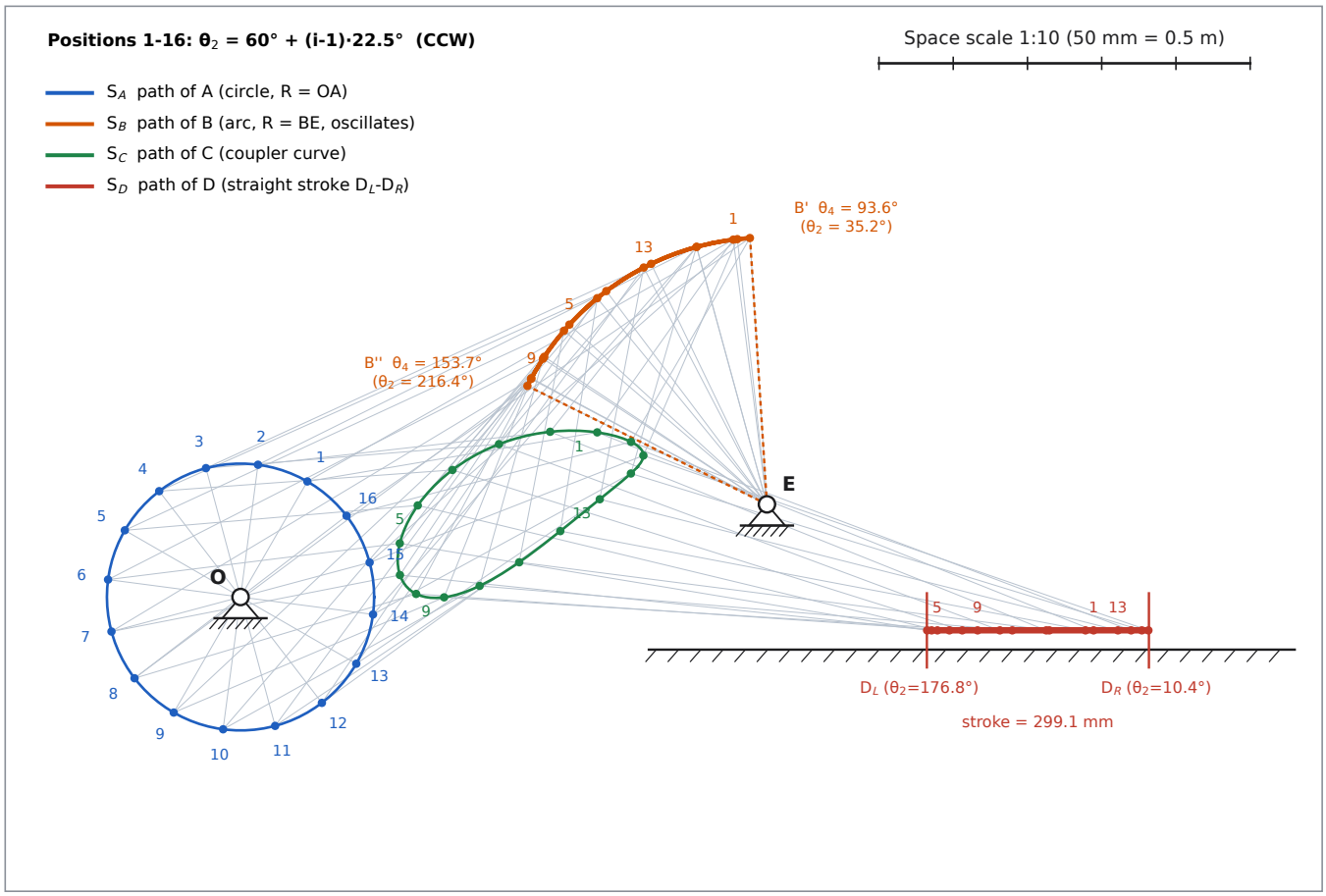


Figure 3 – Sixteen positions of the mechanism, the paths S_A, S_B, S_C, S_D of the moving joints and the limits of motion (scale 1:10).

Pos.	θ_2 (°)	θ_3 (°)	θ_4 (°)	θ_5 (°)	S (mm)	A (x, y) mm	B (x, y) mm	C (x, y) mm	D (x, y) mm
1	60.0	29.63	97.24	−21.78	1,149.0	90.0, 155.9	663.7, 482.2	480.4, 222.1	1,149.0, −45.0
2	82.5	26.44	105.24	−21.84	1,085.3	23.5, 178.5	614.4, 472.4	417.0, 222.9	1,085.3, −45.0
3	105.0	24.68	115.67	−20.42	1,022.8	−46.6, 173.9	553.1, 449.5	348.1, 206.2	1,022.8, −45.0
4	127.5	24.14	126.94	−17.49	972.1	−109.6, 142.8	492.7, 412.7	285.4, 171.4	972.1, −45.0
5	150.0	24.86	137.66	−13.54	938.7	−155.9, 90.0	443.0, 367.5	238.7, 123.6	938.7, −45.0
6	172.5	27.09	146.43	−9.38	924.9	−178.5, 23.5	409.1, 324.1	214.5, 72.4	924.9, −45.0
7	195.0	31.09	151.96	−5.95	930.9	−173.9, −46.6	391.3, 294.3	214.7, 29.6	930.9, −45.0
8	217.5	36.69	153.66	−3.91	954.8	−142.8, −109.6	386.4, 284.8	236.5, 4.1	954.8, −45.0
9	240.0	43.11	151.79	−3.55	992.8	−90.0, −155.9	391.8, 295.2	274.2, −0.4	992.8, −45.0
10	262.5	49.24	146.93	−4.78	1,039.5	−23.5, −178.5	407.4, 321.5	322.0, 15.0	1,039.5, −45.0
11	285.0	53.88	139.41	−7.33	1,089.5	46.6, −173.9	435.7, 359.2	375.4, 46.9	1,089.5, −45.0
12	307.5	55.81	129.42	−10.72	1,138.2	109.6, −142.8	480.5, 403.1	430.7, 88.9	1,138.2, −45.0
13	330.0	54.08	117.45	−14.22	1,181.8	155.9, −90.0	543.1, 444.5	483.9, 131.9	1,181.8, −45.0

Pos.	θ_2 (°)	θ_3 (°)	θ_4 (°)	θ_5 (°)	S (mm)	A (x, y) mm	B (x, y) mm	C (x, y) mm	D (x, y) mm
14	352.5	48.69	105.29	−17.09	1,214.0	178.5, −23.5	614.1, 472.3	525.8, 166.6	1,214.0, −45.0
15	15.0	41.37	96.36	−19.13	1,222.9	173.9, 46.6	669.2, 482.8	542.6, 190.9	1,222.9, −45.0
16	37.5	34.59	93.65	−20.69	1,199.6	142.8, 109.6	686.1, 484.3	526.0, 209.4	1,199.6, −45.0

Table 4 – Sixteen-position analysis of all five moving links (link 2: θ_2; link 3: θ_3; link 4: θ_4; link 5: θ_5; link 6: S). Values are measured from the 1:10 drawing and agree with the analytical solution of Section 5.

Paths of the moving joints.

- **S_A**, the path of A, is a full circle of radius OA = 180 mm about O (pure rotation of link 2).
- **S_B**, the path of B, is a circular arc of radius BE = 360 mm about E. B oscillates between B′ and B″ (link 4 is a rocker).
- **S_C**, the path of C, is a closed oval *coupler curve*, because link 3 has general plane motion.
- **S_D**, the path of D, is a straight segment on y = −H between the limits D_L and D_R (link 6 translates).

Limits of motion.

Link	Quantity	Minimum	Maximum	Notes
2 (crank)	θ_2	0°	360°	full rotation (Grashof crank–rocker)
3 (coupler)	θ_3	24.14°	55.82°	$\omega_3 = 0$ at θ_2 = 126.4° and 308.8°
4 (rocker)	θ_4	93.62° (B′)	153.66° (B″)	swing 60.04°; toggle positions at θ_2 = 35.21° (OA and AB in line, extended) and 216.38° (folded)
5 (rod)	θ_5	−21.98°	−3.48°	$\omega_5 = 0$ at θ_2 = 72.7° and 233.6°
6 (slider)	S	924.5 mm (D_L)	1,223.6 mm (D_R)	stroke 299.1 mm; D_R at θ_2 = 10.40°, D_L at θ_2 = 176.79°

Table 5 – Limits of motion. The extreme positions are where the corresponding velocity vanishes (found exactly by root-finding on the analytical solution).

The slider travels from D_R to D_L (leftwards) while the crank turns 166.4°, and back while the crank turns 193.6°, giving a time ratio Q = 193.6/166.4 = 1.164. The rocker's two strokes take 181.2° and 178.8° of crank rotation, so it is almost symmetrical. Figure 4 shows how the link angles and the slider position vary with crank angle.

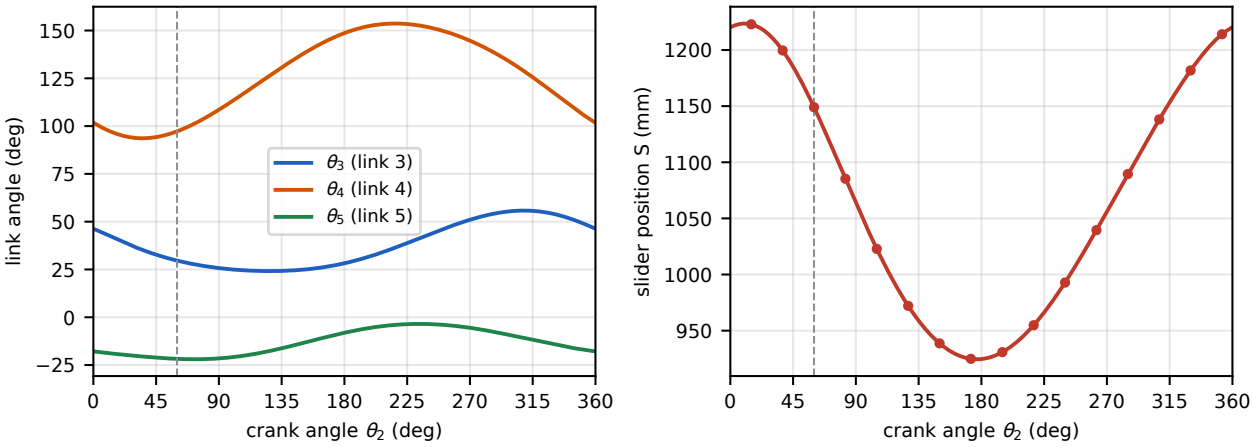


Figure 4 – Link angles θ_3, θ_4, θ_5 and slider position S over one revolution (dots: the 16 positions).

2. Graphical Velocity Analysis ($\theta_2 = 60°$)

Known input. The crank turns about the fixed pivot O, so

$$V_A = \omega_2 \, \overline{OA} = 46.0767 \times 0.180 = 8.294 \text{ m/s},$$

$$\mathbf{V}_A \perp OA, \text{ pointing at } \theta_2 + 90^\circ = 150^\circ \ (\omega_2 \text{ CCW}).$$

Relative-motion equations (course steps, Week 3/9 "Help with the Major Assignment"). Underlining shows what is known: two lines = magnitude and direction known, one line = direction only.

$$\begin{aligned}
&\text{Step 1:} && \underset{\perp BE}{\mathbf{V}_B} = \underset{\perp OA \text{ (known)}}{\mathbf{V}_A} + \underset{\perp AB}{\mathbf{V}_{B/A}} \\
&\text{Step 2:} && \mathbf{V}_C = \mathbf{V}_A + \underset{\perp AC}{\mathbf{V}_{C/A}} = \mathbf{V}_B + \underset{\perp BC}{\mathbf{V}_{C/B}} \\
&\text{Step 3:} && \underset{\parallel x\text{-}x}{\mathbf{V}_D} = \mathbf{V}_C + \underset{\perp CD}{\mathbf{V}_{D/C}} \\
&\text{Step 4:} && \mathbf{V}_{G3} = \mathbf{V}_A + \underset{\perp AG_3}{\mathbf{V}_{G3/A}} = \mathbf{V}_B + \underset{\perp BG_3}{\mathbf{V}_{G3/B}}
\end{aligned}$$

Construction (scale 1 m/s = 20 mm). From the pole *o* draw *oa* = V_A (165.9 mm long, at 150°). Through *a* draw a line ⊥AB, and through *o* a line ⊥BE. They meet at *b* (Step 1). Through *a* draw ⊥AC and through *b* draw ⊥BC; they meet at *c* (Step 2), so the triangle *abc* is the velocity image of link 3. Through *o* draw a horizontal line and through *c* a line ⊥CD; they meet at *d* (Step 3). Point g_3 is the intersection of ⊥AG_3 through *a* and ⊥BG_3 through *b*. Because G_2, G_4 and G_5 are mid-points of OA, EB and CD, the points g_2, g_4, g_5 are the mid-points of *oa*, *ob* and *cd*.

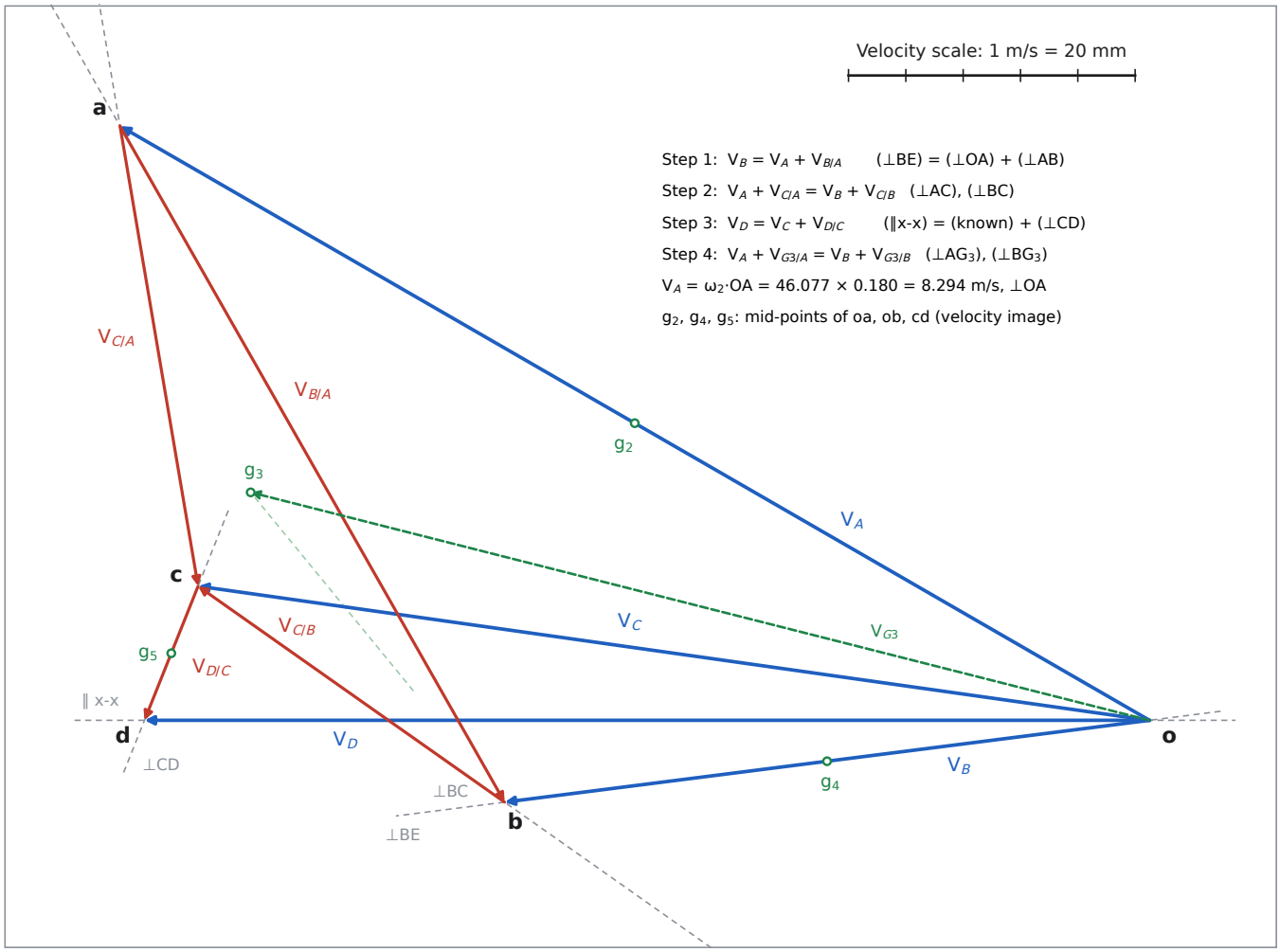


Figure 5 – Velocity vector polygon at θ_2 = 60°, drawn to scale (1 m/s = 20 mm; print at 100 %).

Vector	Direction rule	Length on drawing (mm)	Magnitude (m/s)	Direction (°)
V_A	⊥ OA, = ω2·OA	166.0	8.300	150.0
V_B	⊥ BE	90.5	4.525	187.0
$V_{B/A}$	⊥ AB	108.5	5.425	299.5
V_C	from the image abc	134.0	6.700	172.0
$V_{C/A}$	⊥ AC	65.0	3.250	279.5
$V_{C/B}$	⊥ BC	52.5	2.625	145.0
V_D	‖ x-x (slider path)	140.0	7.000	180.0
$V_{D/C}$	⊥ CD	20.0	1.000	248.0
V_{G2}	mid-point of oa	83.0	4.150	150.0
V_{G3}	from ⊥AG3 & ⊥BG3	129.5	6.475	166.0
V_{G4}	mid-point of ob	45.5	2.275	187.0
V_{G5}	mid-point of cd	137.0	6.850	176.0
V_{G6}	= V_D	140.0	7.000	180.0

Table 6 – Velocities read from the polygon (lengths to the nearest 0.5 mm, angles to the nearest 0.5°, value = length ÷ 20 mm per m/s).

Angular velocities. Each relative velocity is ω × (link length). Its sense follows from the direction of the relative velocity: for example, V_B/A points at 299.5°, which is AB (29.6°) turned *clockwise* by 90°, so link 3 rotates clockwise.

$$\omega_3 = \frac{V_{B/A}}{\overline{AB}} = \frac{5.425}{0.660} = 8.220 \text{ rad/s (CW)},$$

$$\omega_4 = \frac{V_B}{\overline{BE}} = \frac{4.525}{0.360} = 12.569 \text{ rad/s (CCW)},$$

$$\omega_5 = \frac{V_{D/C}}{\overline{CD}} = \frac{1.000}{0.720} = 1.389 \text{ rad/s (CW)},$$

$$\text{check: } \frac{V_{C/A}}{\overline{AC}} = 8.207 \text{ rad/s} \approx \omega_3.$$

Link	ω (rad/s), graphical	Sense
2	46.077 (input)	CCW
3	8.220	CW
4	12.569	CCW
5	1.389	CW
6	0 (translation), V_D = 7.000 m/s to the left	–

Table 7 – Angular velocities from the velocity polygon.

3. Graphical Acceleration Analysis (θ_2 = 60°)

Normal components (calculated first). α_2 = 0, so $a_A^t = 0$ and $a_A = a_A^n$. The other normal components use the velocities read from the polygon (Table 6) and always point from the point towards the centre of the relative rotation.

Component	Formula	Value (m/s²)	Direction
a_A^n	$\omega_2^2\,\overline{OA}$ = 46.077² × 0.180	382.15	A → O (∥AO)
a_B^n	$V_B^2/\overline{BE}$ = 4.525² / 0.360	56.88	B → E (∥BE)
$a_{B/A}^n$	$V_{B/A}^2/\overline{AB}$ = 5.425² / 0.660	44.59	B → A (∥BA)
$a_{C/A}^n$	$V_{C/A}^2/\overline{AC}$ = 3.250² / 0.396	26.67	C → A (∥CA)
$a_{C/B}^n$	$V_{C/B}^2/\overline{BC}$ = 2.625² / 0.3182	21.66	C → B (∥CB)
$a_{D/C}^n$	$V_{D/C}^2/\overline{CD}$ = 1.000² / 0.720	1.39	D → C (∥DC)
$a_{G3/A}^n$	$V_{G3/A}^2/\overline{AG_3}$ = 2.725² / 0.330	22.50	G_3 → A
$a_{G3/B}^n$	$V_{G3/B}^2/\overline{BG_3}$ = 2.800² / 0.3399	23.07	G_3 → B

Table 8 – Normal (centripetal) acceleration components. BC = 318.2 mm and BG_3 = 339.9 mm are measured from the space diagram.

Relative-acceleration equations (course steps).

$$\text{Step 1:}\quad \underset{\parallel BE}{\mathbf{a}_B^n} + \underset{\perp BE}{\mathbf{a}_B^t} = \underset{\parallel AO}{\mathbf{a}_A^n} + \underset{\perp OA\,(=0)}{\mathbf{a}_A^t} + \underset{\parallel BA}{\mathbf{a}_{B/A}^n} + \underset{\perp AB}{\mathbf{a}_{B/A}^t}$$

$$\text{Step 2:}\quad \mathbf{a}_A + \underset{\parallel CA}{\mathbf{a}_{C/A}^n} + \underset{\perp AC}{\mathbf{a}_{C/A}^t} = \mathbf{a}_B + \underset{\parallel CB}{\mathbf{a}_{C/B}^n} + \underset{\perp BC}{\mathbf{a}_{C/B}^t}$$

$$\text{Step 3:}\quad \underset{\parallel x\text{-}x}{\mathbf{a}_D} = \mathbf{a}_C + \underset{\parallel DC}{\mathbf{a}_{D/C}^n} + \underset{\perp CD}{\mathbf{a}_{D/C}^t}$$

$$\text{Step 4:}\quad \mathbf{a}_A + \underset{\perp G_3A}{\mathbf{a}_{G3/A}^t} = \mathbf{a}_B + \mathbf{a}_{G3/B}^n + \underset{\perp G_3B}{\mathbf{a}_{G3/B}^t}$$

Construction (scale 1 mm = 2.5 m/s²). From the pole *o′* draw *o′a′* = a_A (152.9 mm, pointing from A towards O, i.e. at 240°). From *a′* draw $a_{B/A}^n$ (B→A direction) and through its tip a line ⊥AB. From *o′* draw a_B^n (B→E direction) and through its tip a line ⊥BE. The two lines meet at *b′*. From *a′* draw $a_{C/A}^n$ with a line ⊥AC through its tip, and from *b′* draw $a_{C/B}^n$ with a line ⊥BC through its tip; they meet at *c′*. From *c′* draw $a_{D/C}^n$ (only 0.56 mm long) with a line ⊥CD through its tip; this meets the horizontal through *o′* at *d′*. Point g_3' comes from Step 4. Points g_2' and g_4' are the mid-points of *o′a′* and *o′b′*, and g_5' is the mid-point of *c′d′*.

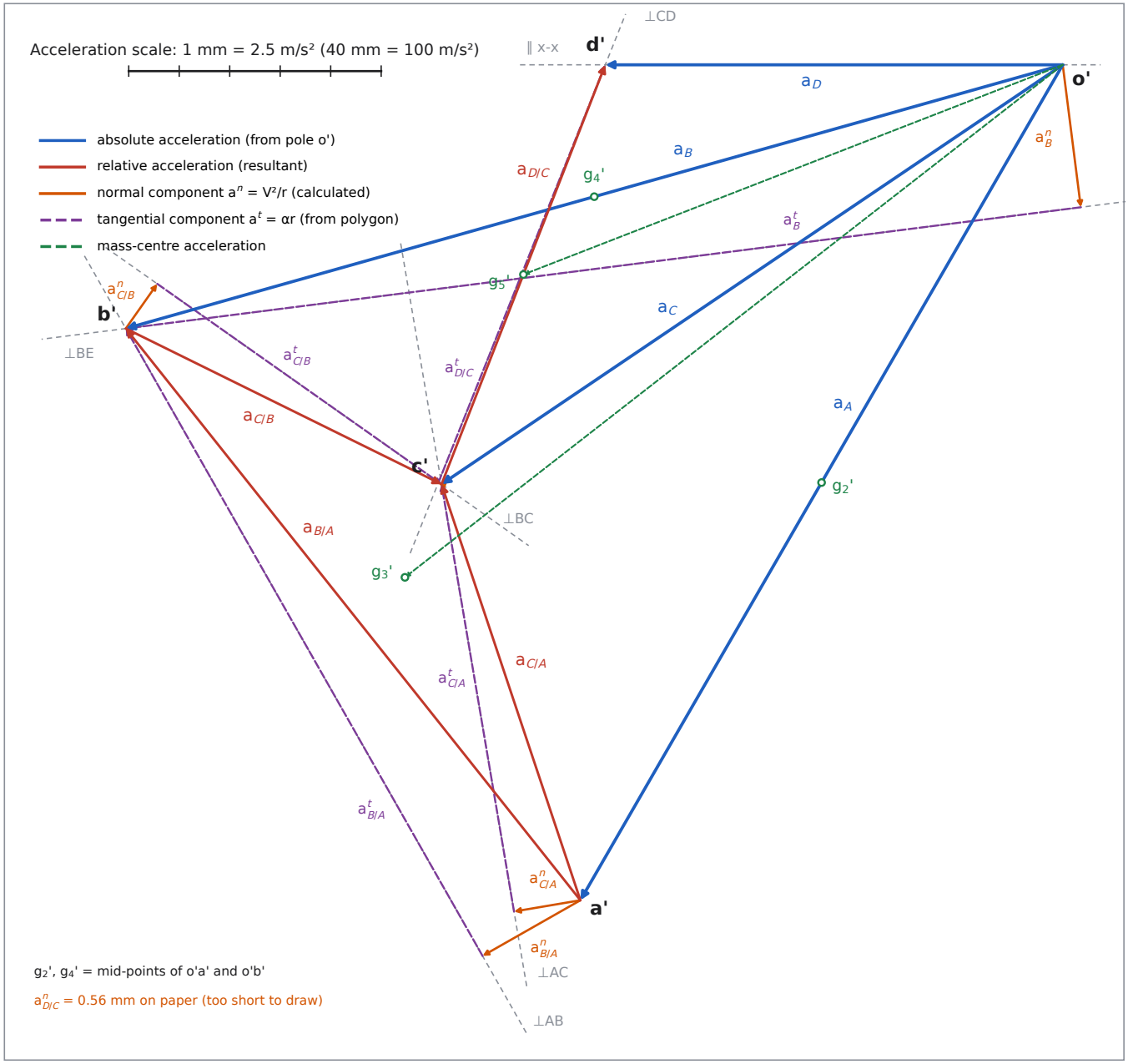


Figure 6 – Acceleration vector polygon at $\theta_2 = 60°$, drawn to scale (1 mm = 2.5 m/s²; print at 100 %).

Vector	Construction	Length (mm)	Magnitude (m/s²)	Direction (°)
a_A	= a_A^n, ∥AO towards O (a_A^t = 0)	153.0	382.5	240.0
a_B^n	∥BE towards E, = V_B²/BE	23.0	57.5	277.0
a_B^t	⊥BE	152.5	381.2	187.0
a_B	resultant	154.0	385.0	195.5
$a_{B/A}^n$	∥BA towards A, = V_B/A²/AB	18.0	45.0	209.5
$a_{B/A}^t$	⊥AB	114.5	286.2	119.5
$a_{B/A}$	resultant	115.5	288.8	128.5
$a_{C/A}^n$	∥CA towards A, = V_C/A²/AC	10.5	26.2	189.5
$a_{C/A}^t$	⊥AC	68.5	171.2	99.5
$a_{C/B}^n$	∥CB towards B, = V_C/B²/BC	8.5	21.2	55.0
$a_{C/B}^t$	⊥BC	55.0	137.5	325.0
a_C	resultant	119.0	297.5	214.0

Vector	Construction	Length (mm)	Magnitude (m/s²)	Direction (°)
$a^n_{D/C}$	∥DC towards C, = V_D/C²/CD	0.5	1.2	158.0
$a^t_{D/C}$	⊥CD	71.5	178.8	68.0
a_D	∥x-x	72.5	181.2	180.0
a_{G2}	mid-point of o'a'	76.5	191.2	240.0
a_{G3}	step 4	132.0	330.0	218.0
a_{G4}	mid-point of o'b'	77.0	192.5	195.5
a_{G5}	mid-point of c'd'	91.5	228.8	201.0
a_{G6}	= a_D	72.5	181.2	180.0

Table 9 – Accelerations read from the polygon (lengths to the nearest 0.5 mm; value = length × 2.5 m/s² per mm).

Angular accelerations come from the tangential components ($a^t = \alpha\, r$), with the sense taken from the polygon:

$$\alpha_3 = \frac{a^t_{B/A}}{\overline{AB}} = \frac{286.2}{0.660} = 433.7\ \text{rad/s}^2\ (\text{CCW}),$$

$$\alpha_4 = \frac{a^t_{B}}{\overline{BE}} = \frac{381.2}{0.360} = 1,059.0\ \text{rad/s}^2\ (\text{CCW}),$$

$$\alpha_5 = \frac{a^t_{D/C}}{\overline{CD}} = \frac{178.8}{0.720} = 248.3\ \text{rad/s}^2\ (\text{CCW}),$$

$$\text{check: } \frac{a^t_{C/A}}{\overline{AC}} = 432.4\ \text{rad/s}^2 \approx \alpha_3.$$

Quantity	Graphical value	Direction
a_A	382.5 m/s²	240.0°
a_B	385.0 m/s²	195.5°
a_C	297.5 m/s²	214.0°
a_D	181.2 m/s²	180.0° (to the left)
$a_{G2}, a_{G3}, a_{G4}, a_{G5}, a_{G6}$	191.2, 330.0, 192.5, 228.8, 181.2 m/s²	240.0°, 218.0°, 195.5°, 201.0°, 180.0°
$\alpha_3, \alpha_4, \alpha_5$	433.7, 1,059.0, 248.3 rad/s²	all CCW

Table 10 – Summary of the graphical acceleration analysis.

4. Instantaneous Centres and Velocities (Kennedy's Theorem)

Number of ICs (Kennedy Part 1). With n = 6 links, $C = \frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15.$

Known ICs by inspection (7). The pins give I_{12} = O, I_{23} = A, I_{34} = B, I_{14} = E, I_{35} = C and I_{56} = D. The slider gives I_{16}, at infinity perpendicular to the horizontal guide (any vertical line passes through I_{16}).

Unknown ICs (8). I_{13}, I_{24}, I_{15}, I_{36}, I_{25}, I_{45}, I_{26} and I_{46}.

Kennedy circle (Parts 2 and 3). Six points numbered 1–6 are placed on a circle. Known ICs are drawn as solid sides and unknown ones as dashed sides. An unknown side can be found once it closes two triangles whose other four sides are known: by Kennedy's theorem the unknown IC lies on both corresponding lines, so it is their intersection.

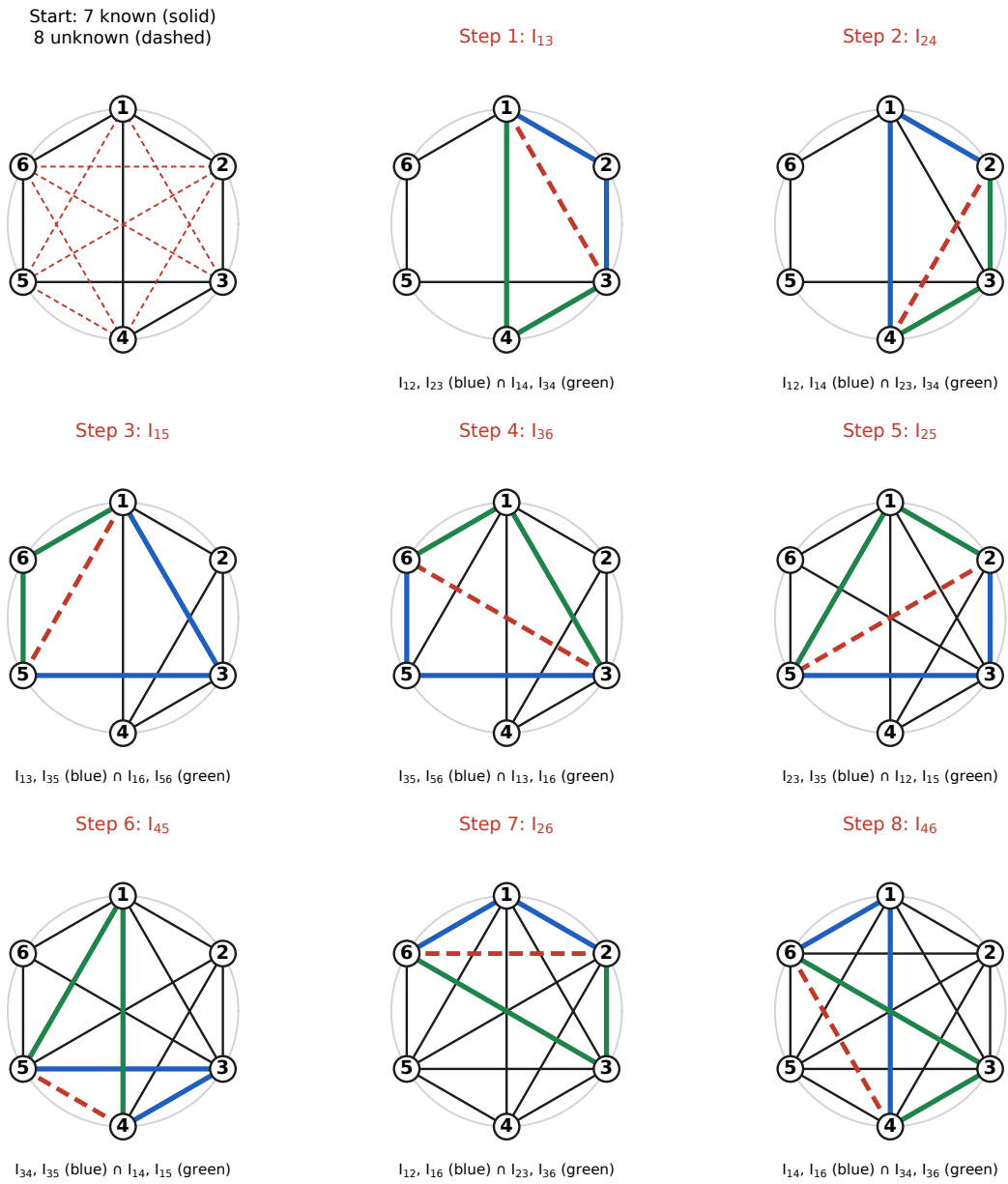


Figure 7 – Kennedy circle, step by step: in each step the dashed red side is closed by the blue and the green triangles.

Step	Unknown IC	Line 1 (through)	Line 2 (through)	Location found (x, y) mm
1	I13	I_{12} (O), I_{23} (A) – line OA	I_{14} (E), I_{34} (B) – line EB	594.2, 1,029.3

Step	Unknown IC	Line 1 (through)	Line 2 (through)	Location found (x, y) mm
2	I24	I_{12} (O), I_{14} (E) – line OE	I_{23} (A), I_{34} (B) – line AB	−266.8, −47.1
3	I15	I_{13}, I_{35} (C)	I_{16} (∞ vertical), I_{56} (D) – vertical through D	1,149.0, 4,963.7
4	I36	I_{35} (C), I_{56} (D) – line CD	I_{13}, I_{16} – vertical through I_{13}	594.2, 176.6
5	I25	I_{23} (A), I_{35} (C) – line AC	I_{12} (O), I_{15}	33.9, 146.4
6	I45	I_{34} (B), I_{35} (C) – line BC	I_{14} (E), I_{15}	753.1, 608.9
7	I26	I_{12} (O), I_{16} – vertical through O	I_{23} (A), I_{36}	0.0, 152.2
8	I46	I_{14} (E), I_{16} – vertical through E	I_{34} (B), I_{36}	709.1, 681.6

Table 11 – Location of the unknown ICs. A third Kennedy line through each IC was used as a check; all were collinear to better than 10^{-10} m.

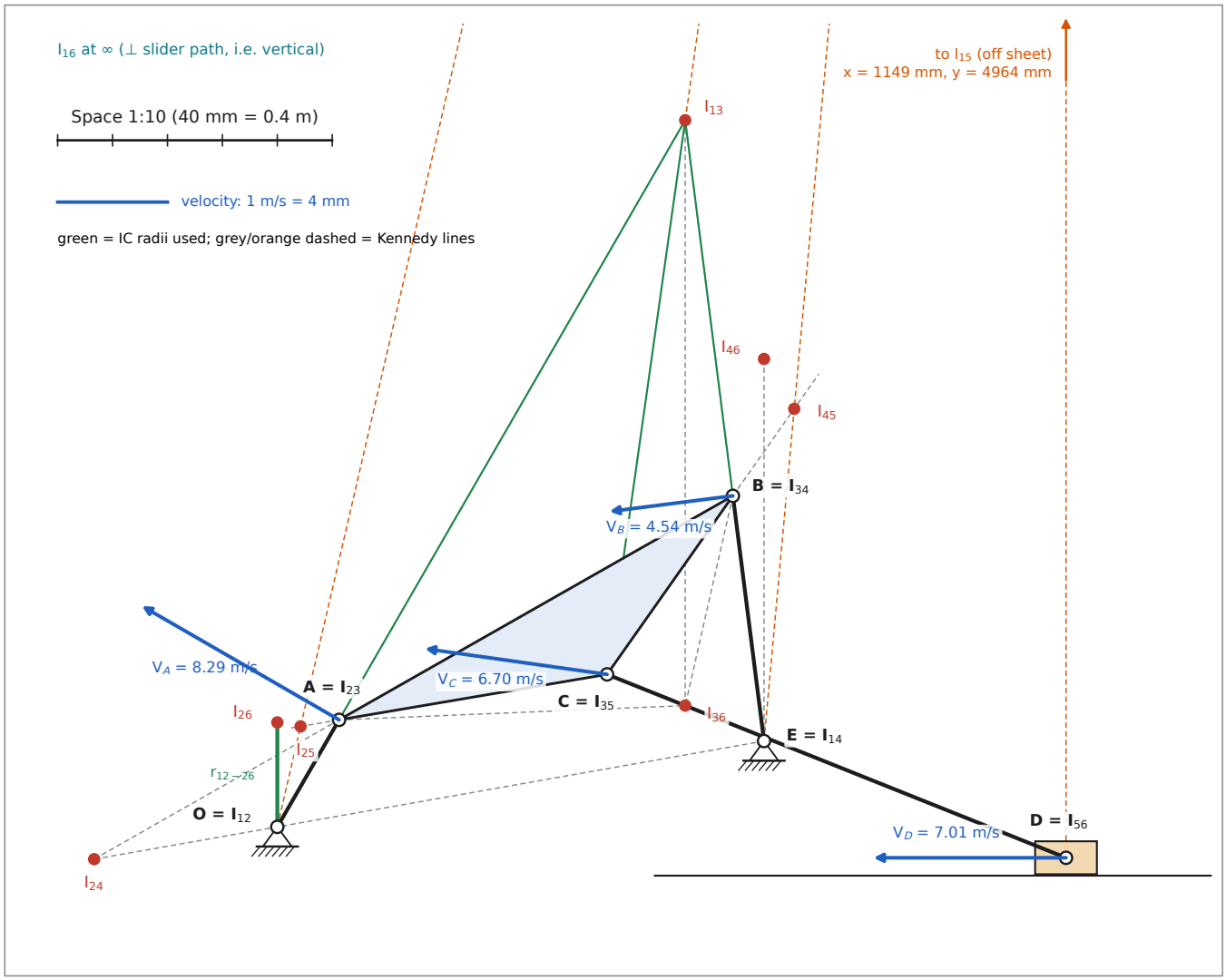


Figure 8 – All instantaneous centres and the velocities of A, B, C, D drawn to scale on the mechanism (space 1:10; velocity 1 m/s = 4 mm). I_{15} lies off the sheet.

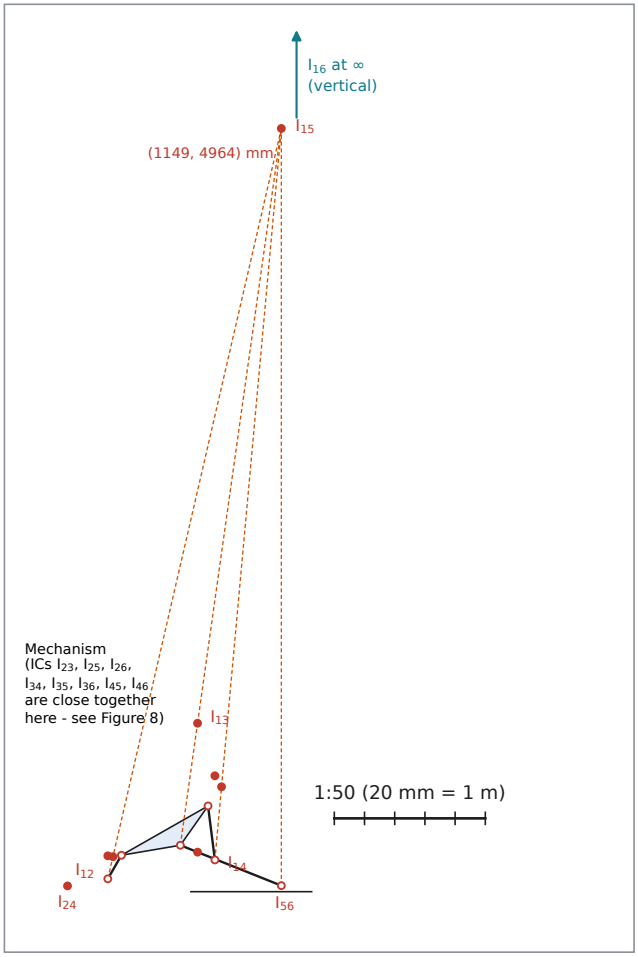


Figure 8b – Overview of all 15 instantaneous centres at scale 1:50, including I_{15} (about 5 m above the slider), with the Kennedy lines that locate it; I_{16} is at infinity in the vertical direction.

Velocities from the ICs (radii read from the 1:10 drawing to the nearest 0.5 mm, i.e. 5 mm full size):

Step	Equation	Radius (mm, full size)	Result
1	$V_A = \omega_2 \, r_{I_{12}A}$	180	V_A = 8.294 m/s ⊥ I_{12}A
2	$\omega_3 = V_A / r_{I_{13}A}$ (link 3 turns about I_{13})	1,010	ω_3 = 8.212 rad/s CW
3	$V_B = \omega_3 \, r_{I_{13}B}$	550	V_B = 4.516 m/s ⊥ I_{13}B
4	$V_C = \omega_3 \, r_{I_{13}C}$	815	V_C = 6.693 m/s ⊥ I_{13}C
5	$\omega_4 = V_B / r_{I_{14}B}$	360	ω_4 = 12.546 rad/s CCW
6	$V_D = \omega_2 \, r_{I_{12}I_{26}}$ (I_{26} is a point of link 2 moving with the slider)	150	V_D = 6.912 m/s ←
7	$\omega_5 = V_C / r_{I_{15}C}$, $V_D = \omega_5 \, r_{I_{15}D}$ (I_{15} off-sheet, coordinates from the Kennedy lines)	4,788, 5,009	ω_5 = 1.398 rad/s CW; V_D = 7.000 m/s

Table 12 – Velocities of A, B, C and D using the instantaneous centres.

The IC results agree with the velocity polygon and the analytical solution (Table 16). The largest deviation (V_D via I_{26}, −1.4 %) comes from reading the short radius O–I_{26} (exact value 152.2 mm) to ±5 mm at 1:10 scale. Using I_{15} (step 7),

or drawing that region at a larger scale, removes the error. The angular-velocity-ratio theorem gives a neat check: I_{24} lies on the line of centres OE, so $\omega_4/\omega_2 = I_{12}I_{24} / I_{14}I_{24} = 0.2734 \rightarrow \omega_4 = 12.599$ rad/s.

5. Analytical Solution — Vector Loops and Complex Numbers

The course template "Approach for the analytical part of the major assignment" (Week 7) splits the mechanism into Part 1 (crank, point A), Part 2 (coupler and rocker, points B and C) and Part 3 (slider, point D).

Vectors. $\mathbf{R}_2 = l_2 e^{j\theta_2}$ (O→A), $\mathbf{R}_3 = l_3 e^{j\theta_3}$ (A→B), $\mathbf{R}_4 = l_4 e^{j\theta_4}$ (E→B), $\mathbf{R}_1 = l_1 e^{j\theta_1}$ (O→E), $\mathbf{R}_{AC} = \overline{AC}\, e^{j(\theta_3-\beta)}$ (A→C; C lies below AB), $\mathbf{R}_5 = l_5 e^{j\theta_5}$ (C→D) and $\mathbf{R}_D = S - jH$ (O→D).

$$\text{Loop 1 (O-A-B-E):}\quad l_2 e^{j\theta_2} + l_3 e^{j\theta_3} = l_1 e^{j\theta_1} + l_4 e^{j\theta_4}$$

$$\text{Loop 2 (O-A-C-D):}\quad l_2 e^{j\theta_2} + \overline{AC}\, e^{j(\theta_3-\beta)} + l_5 e^{j\theta_5} = S - jH$$

5.1 Position analysis

Part 1 – crank (point A).

$$x_A = l_2 \cos\theta_2 = 180\cos 60^\circ = 90.00 \text{ mm}, \qquad y_A = l_2 \sin\theta_2 = 155.88 \text{ mm}.$$

Part 2 – coupler and rocker (points B and C). Following the template, let $\mathbf{R}_{AE} = \mathbf{R}_1 - \mathbf{R}_2 = \mathrm{Re} + j\,\mathrm{Im}$:

$$\mathrm{Re} = l_1\cos\theta_1 - l_2\cos\theta_2 = 0.720\cos 10^\circ - 0.180\cos 60^\circ = 0.61906 \text{ m}$$

$$\mathrm{Im} = l_1\sin\theta_1 - l_2\sin\theta_2 = 0.720\sin 10^\circ - 0.180\sin 60^\circ = -0.03086 \text{ m}$$

Loop 1 becomes $l_3 e^{j\theta_3} = \mathrm{Re} + j\mathrm{Im} + l_4 e^{j\theta_4}$. Taking the squared magnitude of both sides eliminates θ_3:

$$\mathrm{Re}\cos\theta_4 + \mathrm{Im}\sin\theta_4 + C = 0, \qquad C = \frac{\mathrm{Re}^2 + \mathrm{Im}^2 + l_4^2 - l_3^2}{2l_4} = 0.10860 \text{ m}.$$

With $\sin\theta_4 = \dfrac{2t}{1+t^2}$, $\cos\theta_4 = \dfrac{1-t^2}{1+t^2}$ and $t = \tan(\theta_4/2)$:

$$(C - \mathrm{Re})\,t^2 + 2\,\mathrm{Im}\,t + (C + \mathrm{Re}) = 0,$$

$$t = \frac{-\mathrm{Im} \pm \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2 - C^2}}{C - \mathrm{Re}} = \frac{0.03086 \pm 0.61024}{-0.51047}.$$

The two roots are $t = 1.1350$ (θ_4 = 97.237°) and $t = -1.2559$ (θ_4 = −102.944°). The template takes the solution with $0 < \theta_4 < \pi$, which is the open configuration of the handout figure (B above E):

$$\theta_4 = 97.237^\circ \quad (\text{the "}-\text{" sign here}).$$

From the triangle AB′B of the template:

$$\theta_3 = 2\arctan\frac{\mathrm{Im} + l_4\sin\theta_4}{l_3 + \mathrm{Re} + l_4\cos\theta_4} = 2\arctan\frac{0.32627}{1.23371} = 29.627^\circ.$$

Hence $\mathbf{R}_B = \mathbf{R}_1 + l_4 e^{j\theta_4}$, i.e. B = (663.71, 482.16) mm, and $\mathbf{R}_C = \mathbf{R}_2 + \overline{AC} e^{j(\theta_3-\beta)}$ with $\theta_3 - \beta$ = 9.627°, i.e. C = (480.42, 222.11) mm. Check: |B − A| = 660.000 mm ✓.

Part 3 – slider (point D). The imaginary part of loop 2 is $y_C + l_5\sin\theta_5 = -H$ (the slider runs H *below* O, so $y_D = -H$):

$$\theta_5 = \arcsin\!\left(\frac{-H - y_C}{l_5}\right) = \arcsin\!\left(\frac{-45.0 - 222.11}{720}\right) = -21.776^\circ,$$

$$S = x_D = x_C + l_5\cos\theta_5 = 480.42 + 720\cos(-21.776^\circ) = 1,149.04\ \text{mm}.$$

Mass-centre positions. $\mathbf{R}_{G2} = \frac{l_2}{2}e^{j\theta_2}$, $\mathbf{R}_{G3} = \mathbf{R}_A + \frac{l_3}{2}e^{j(\theta_3-\beta/2)}$, $\mathbf{R}_{G4} = \mathbf{R}_1 + \frac{l_4}{2}e^{j\theta_4}$, $\mathbf{R}_{G5} = \mathbf{R}_C + \frac{l_5}{2}e^{j\theta_5}$, $\mathbf{R}_{G6} = \mathbf{R}_D$.

5.2 Velocity analysis

Differentiating loop 1 with respect to time (θ₁, l's constant):

$$jl_2\omega_2 e^{j\theta_2} + jl_3\omega_3 e^{j\theta_3} - jl_4\omega_4 e^{j\theta_4} = 0 \quad\Leftrightarrow\quad \mathbf{V}_A + \mathbf{V}_{B/A} = \mathbf{V}_B.$$

The real and imaginary parts give two linear equations in ω₃, ω₄. Solving them (Week 7):

$$\omega_3 = \frac{l_2\omega_2\sin(\theta_4-\theta_2)}{l_3\sin(\theta_3-\theta_4)} = \frac{0.180(46.0767)\sin(37.237^\circ)}{0.660\sin(-67.610^\circ)} = -8.2241\ \text{rad/s (CW)}$$

$$\omega_4 = \frac{l_2\omega_2\sin(\theta_2-\theta_3)}{l_4\sin(\theta_4-\theta_3)} = \frac{0.180(46.0767)\sin(30.373^\circ)}{0.360\sin(67.610^\circ)} = 12.5985\ \text{rad/s (CCW)}$$

Point velocities, using $je^{j\theta} = -\sin\theta + j\cos\theta$:

$$\mathbf{V}_A = l_2\omega_2(-\sin\theta_2 + j\cos\theta_2) = -7.1826 + j\,4.1469\ \text{m/s}$$

$$\mathbf{V}_B = l_4\omega_4(-\sin\theta_4 + j\cos\theta_4) = -4.4993 + j\,(-0.5713)\ \text{m/s}$$

$$\mathbf{V}_C = \mathbf{V}_A + \overline{AC}\,\omega_3\,[-\sin(\theta_3-\beta) + j\cos(\theta_3-\beta)] = -6.6380 + j\,0.9360\ \text{m/s}$$

Differentiating loop 2, $\mathbf{V}_C + jl_5\omega_5 e^{j\theta_5} = \dot{S}$ (real, since the slider moves horizontally):

$$\text{Im: } V_{Cy} + l_5\omega_5\cos\theta_5 = 0 \;\Rightarrow\; \omega_5 = \frac{-V_{Cy}}{l_5\cos\theta_5} = \frac{-0.9360}{0.720\cos(-21.776^\circ)} = -1.4000\ \text{rad/s (CW)}$$

$$\text{Re: } V_D = \dot{S} = V_{Cx} - l_5\omega_5\sin\theta_5 = -7.0119\ \text{m/s (to the left)}$$

Mass-centre velocities: $\mathbf{V}_{G3} = \mathbf{V}_A + j\frac{l_3}{2}\omega_3 e^{j(\theta_3-\beta/2)} = -6.2710 + j\,1.5906$ m/s; $\mathbf{V}_{G5} = \mathbf{V}_C + j\frac{l_5}{2}\omega_5 e^{j\theta_5} = -6.8250 + j\,0.4680$ m/s; $\mathbf{V}_{G2} = \tfrac{1}{2}\mathbf{V}_A$, $\mathbf{V}_{G4} = \tfrac{1}{2}\mathbf{V}_B$.

5.3 Acceleration analysis

Differentiating the velocity loop again:

$$(j\alpha_2 - \omega_2^2)l_2e^{j\theta_2} + (j\alpha_3 - \omega_3^2)l_3e^{j\theta_3} - (j\alpha_4 - \omega_4^2)l_4e^{j\theta_4} = 0 \quad\Leftrightarrow\quad \mathbf{a}_A + \mathbf{a}_{B/A} = \mathbf{a}_B.$$

Separating real and imaginary parts and using the Week 7 constants (with a = l_2, b = l_3, c = l_4):

$$A = c\sin\theta_4 = 0.35713, \qquad B = b\sin\theta_3 = 0.32627,$$
$$D = c\cos\theta_4 = -0.04535, \quad E = b\cos\theta_3 = 0.57371,$$

$$C = a\alpha_2\sin\theta_2 + a\omega_2^2\cos\theta_2 + b\omega_3^2\cos\theta_3 - c\omega_4^2\cos\theta_4 = 237.077,$$

$$F = a\alpha_2\cos\theta_2 - a\omega_2^2\sin\theta_2 - b\omega_3^2\sin\theta_3 + c\omega_4^2\sin\theta_4 = -296.335,$$

$$\alpha_3 = \frac{CD - AF}{AE - BD} = 432.794 \text{ rad/s}^2 \text{ (CCW)},$$

$$\alpha_4 = \frac{CE - BF}{AE - BD} = 1,059.233 \text{ rad/s}^2 \text{ (CCW)}.$$

Point accelerations:

$$\mathbf{a}_A = (j\alpha_2 - \omega_2^2) l_2 e^{j\theta_2} = -191.076 + j(-330.953) \text{ m/s}^2$$

$$\mathbf{a}_B = (j\alpha_4 - \omega_4^2) l_4 e^{j\theta_4} = -371.088 + j(-104.721) \text{ m/s}^2$$

$$\mathbf{a}_C = \mathbf{a}_A + (j\alpha_3 - \omega_3^2)\overline{AC}\, e^{j(\theta_3 - \beta)} = -246.144 + j(-166.459) \text{ m/s}^2$$

Differentiating loop 2 again, $\mathbf{a}_C + (j\alpha_5 - \omega_5^2) l_5 e^{j\theta_5} = \ddot{S}$:

$$\alpha_5 = \frac{l_5\omega_5^2 \sin\theta_5 - a_{Cy}}{l_5 \cos\theta_5} = 248.176 \text{ rad/s}^2 \text{ (CCW)},$$

$$a_D = \ddot{S} = a_{Cx} - l_5\alpha_5 \sin\theta_5 - l_5\omega_5^2 \cos\theta_5 = -181.164 \text{ m/s}^2.$$

Mass-centre accelerations, used in Section 6:

Link	$\mathbf{a}_G$ (m/s²)	$\lvert a_G \rvert$ (m/s²)	Direction (°)	α (rad/s²)
2	−95.538 + j(−165.476)	191.076	240.00	0
3	−260.072 + j(−203.926)	330.490	218.10	432.794
4	−185.544 + j(−52.361)	192.791	195.76	1,059.233
5	−213.654 + j(−83.230)	229.293	201.28	248.176
6	−181.164 + j0	181.164	180.00	0

Table 13 – Mass-centre accelerations at θ₂ = 60° (analytical).

5.4 Results and comparison

Link / point	Position	Velocity	Acceleration
2 (A)	θ₂ = 60°, A = (90.00, 155.88) mm	ω₂ = 46.077 rad/s CCW; V_A = 8.294 m/s ∠150.0°	α₂ = 0; a_A = 382.15 m/s² ∠240.0°
3 (B, C)	θ₃ = 29.627°, C = (480.42, 222.11) mm	ω₃ = −8.2241 rad/s; V_C = 6.704 m/s ∠172.0°	α₃ = 432.79 rad/s²; a_C = 297.15 m/s² ∠214.1°
4 (B)	θ₄ = 97.237°, B = (663.71, 482.16) mm	ω₄ = 12.5985 rad/s; V_B = 4.535 m/s ∠187.2°	α₄ = 1,059.23 rad/s²; a_B = 385.58 m/s² ∠195.8°
5 (C→D)	θ₅ = −21.776°	ω₅ = −1.4000 rad/s	α₅ = 248.18 rad/s²
6 (D)	S = 1,149.04 mm, y = −45.0 mm	V_D = −7.0119 m/s	a_D = −181.164 m/s²

Table 14 – Analytical results at θ₂ = 60° (signed angular quantities: + = CCW).

Pos.	θ₂ (°)	θ₃ (°)	θ₄ (°)	θ₅ (°)	S (mm)	ω₃	ω₄	ω₅	V_D (m/s)	α₃	α₄	α₅	a_D (m/s²)
1	60.0	29.627	97.237	−21.776	1,149.04	−8.224	12.599	−1.400	−7.012	432.8	1,059.2	248.2	−181.2
2	82.5	26.444	105.238	−21.844	1,085.30	−4.952	19.484	1.296	−7.656	338.2	574.9	368.8	23.1

Pos.	θ_2 (°)	θ_3 (°)	θ_4 (°)	θ_5 (°)	S (mm)	ω_3	ω_4	ω_5	V_D (m/s)	α_3	α_4	α_5	a_D (m/s²)
3	105.0	24.684	115.672	−20.420	1,022.85	−2.327	22.713	4.543	−6.795	287.6	197.4	371.4	166.9
4	127.5	24.143	126.943	−17.492	972.10	0.125	22.986	7.280	−5.008	298.2	−131.4	253.3	242.5
5	150.0	24.862	137.663	−13.539	938.68	2.913	20.437	8.619	−2.792	362.9	−470.3	50.9	272.0
6	172.5	27.094	146.429	−9.384	924.87	6.335	15.004	8.071	−0.445	433.2	−791.5	−177.4	274.8
7	195.0	31.092	151.961	−5.947	930.86	9.992	7.440	5.765	1.815	398.2	−941.9	−347.0	250.3
8	217.5	36.690	153.660	−3.914	954.84	12.656	−0.366	2.477	3.731	204.8	−861.4	−407.1	194.7
9	240.0	43.112	151.794	−3.548	992.84	13.259	−7.065	−0.949	5.079	−69.0	−715.2	−387.6	119.9
10	262.5	49.243	146.927	−4.779	1,039.53	11.438	−12.749	−4.000	5.770	−359.3	−631.8	−321.1	43.4
11	285.0	53.877	139.412	−7.330	1,089.48	7.123	−17.990	−6.286	5.861	−652.2	−600.3	−206.0	−18.8
12	307.5	55.809	129.420	−10.717	1,138.16	0.439	−22.798	−7.336	5.496	−898.7	−500.1	−30.7	−66.7
13	330.0	54.079	117.454	−14.222	1,181.81	−7.562	−25.634	−6.737	4.633	−918.2	−89.6	164.4	−148.6
14	352.5	48.693	105.290	−17.093	1,214.03	−13.877	−22.930	−4.944	2.675	−485.2	783.7	218.1	−324.2
15	15.0	41.372	96.364	−19.125	1,222.90	−15.168	−12.494	−3.554	−0.796	163.3	1,566.8	101.3	−462.2
16	37.5	34.594	93.655	−20.687	1,199.60	−12.168	1.362	−2.846	−4.559	463.6	1,551.4	100.5	−386.1

Table 15 – Analytical solution at the 16 positions (ω in rad/s, α in rad/s²; + = CCW, V_D and a_D + = to the right).

Quantity	Unit	Graphical	IC method	Analytical	Computational	Graph. vs anal.	IC vs anal.
V_A	m/s	8.300	8.294	8.294	8.294	0.07 %	0.00 %
V_B	m/s	4.525	4.516	4.535	4.535	−0.23 %	−0.42 %
V_C	m/s	6.700	6.693	6.704	6.704	−0.05 %	−0.17 %
V_D	m/s	7.000	6.912	7.012	7.012	−0.17 %	−1.43 %
ω_3	rad/s	8.220	8.212	8.224	8.224	−0.05 %	−0.15 %
ω_4	rad/s	12.569	12.546	12.599	12.599	−0.23 %	−0.42 %
ω_5	rad/s	1.389	1.398	1.400	1.400	−0.79 %	−0.17 %
a_B	m/s²	385.000	–	385.581	385.581	−0.15 %	–
a_C	m/s²	297.500	–	297.146	297.146	0.12 %	–
a_D	m/s²	181.250	–	181.164	181.164	0.05 %	–
α_3	rad/s²	433.712	–	432.794	432.794	0.21 %	–
α_4	rad/s²	1,059.028	–	1,059.233	1,059.233	−0.02 %	–
α_5	rad/s²	248.264	–	248.176	248.176	0.04 %	–
a_{G2}	m/s²	191.250	–	191.076	191.076	0.09 %	–
a_{G3}	m/s²	330.000	–	330.490	330.490	−0.15 %	–
a_{G4}	m/s²	192.500	–	192.791	192.791	−0.15 %	–
a_{G5}	m/s²	228.750	–	229.293	229.293	−0.24 %	–

Table 16 – Comparison of all methods at θ_2 = 60° (magnitudes). The computational model differs from the analytical solution by less than 10^{-7} in every quantity (Section 7).

Correlation. The graphical results agree with the analytical ones to within 0.79 % for velocities and 0.24 % for accelerations. The remaining differences come only from reading the drawings to ±0.5 mm and ±0.5°. Short vectors suffer most: V_D/C is just 20 mm long, so a 0.25 mm reading error is already 1.2 % of it, which is why ω_5 shows the largest graphical error. Choosing a larger scale for short vectors, or computing them analytically, improves the result. The analytical method is exact for the model, and its numbers were also verified against numerical differentiation of the position solution.

6. Dynamic Force Analysis — Matrix Method (θ_2 = 60°)

Assumptions. Rigid links; frictionless pins; constant crank speed (α_2 = 0); no external load on the slider; gravity neglected (at θ_2 = 60° the inertia forces are 18–34 times the link weights; the effect of gravity is quantified in Section 7). The slider guide is frictionless in the baseline solution. Coulomb friction μ = 0.3, as in the lecture help example, is solved as a variant.

Notation (Week 9). $\mathbf{F}_{ij}$ is the force of link *i* on link *j*, and $\mathbf{F}_{ji} = -\mathbf{F}_{ij}$. $\mathbf{R}_{ij}$ is the vector from the mass centre G_j to the joint where $\mathbf{F}_{ij}$ acts. The cross product of two planar vectors is $\mathbf{R} \times \mathbf{F} = R_x F_y - R_y F_x$.

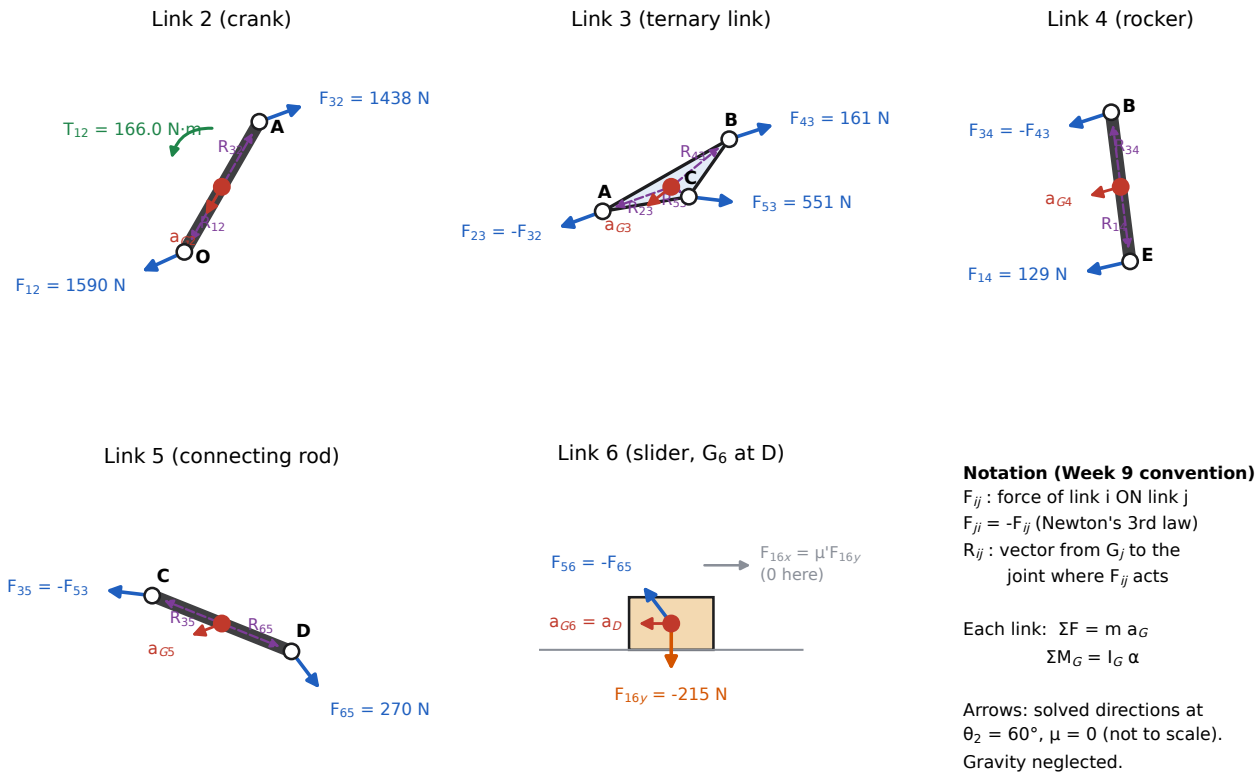


Figure 9 – Free-body diagrams of links 2–6 at θ_2 = 60°, showing the solved force directions (not to force scale).

Vector	x (m)	y (m)	Vector	x (m)	y (m)
R_{12} ($G_2 \to O$)	−0.04500	−0.07794	R_{32} ($G_2 \to A$)	0.04500	0.07794
R_{23} ($G_3 \to A$)	−0.31083	−0.11085	R_{43} ($G_3 \to B$)	0.26289	0.21543
R_{53} ($G_3 \to C$)	0.07960	−0.04462	R_{34} ($G_4 \to B$)	−0.02267	0.17857
R_{14} ($G_4 \to E$)	0.02267	−0.17857	R_{35} ($G_5 \to C$)	−0.33431	0.13356
R_{65} ($G_5 \to D$)	0.33431	−0.13356			

Table 17 – Position vectors from each mass centre to its joints at θ_2 = 60°.

Equations of motion (14 equations, 14 unknowns $F_{12x}, F_{12y}, F_{32x}, F_{32y}, F_{43x}, F_{43y}, F_{53x}, F_{53y}, F_{14x}, F_{14y}, F_{65x}, F_{65y}, F_{16y}, T_{12}$):

Link 2:

$$(1)\quad F_{12x}+F_{32x}=m_2a_{G2x} \qquad (2)\quad F_{12y}+F_{32y}=m_2a_{G2y}$$

$$(3)\quad T_{12}+(R_{12x}F_{12y}-R_{12y}F_{12x})+(R_{32x}F_{32y}-R_{32y}F_{32x})=I_{G2}\alpha_2$$

Link 3: $(\mathbf{F}_{23}=-\mathbf{F}_{32})$

$$(4)\quad -F_{32x}+F_{43x}+F_{53x}=m_3a_{G3x} \qquad (5)\quad -F_{32y}+F_{43y}+F_{53y}=m_3a_{G3y}$$

$$(6)\quad -(R_{23x}F_{32y}-R_{23y}F_{32x})+(R_{43x}F_{43y}-R_{43y}F_{43x})+(R_{53x}F_{53y}-R_{53y}F_{53x})=I_{G3}\alpha_3$$

Link 4: $(\mathbf{F}_{34}=-\mathbf{F}_{43})$

$$(7)\quad -F_{43x}+F_{14x}=m_4a_{G4x} \qquad (8)\quad -F_{43y}+F_{14y}=m_4a_{G4y}$$

$$(9)\quad -(R_{34x}F_{43y}-R_{34y}F_{43x})+(R_{14x}F_{14y}-R_{14y}F_{14x})=I_{G4}\alpha_4$$

Link 5: $(\mathbf{F}_{35}=-\mathbf{F}_{53})$

$$(10)\quad -F_{53x}+F_{65x}=m_5a_{G5x} \qquad (11)\quad -F_{53y}+F_{65y}=m_5a_{G5y}$$

$$(12)\quad -(R_{35x}F_{53y}-R_{35y}F_{53x})+(R_{65x}F_{65y}-R_{65y}F_{65x})=I_{G5}\alpha_5$$

Link 6: $(\mathbf{F}_{56}=-\mathbf{F}_{65},\ F_{16x}=\mu' F_{16y})$

$$(13)\quad -F_{65x}+\mu' F_{16y}=m_6a_{G6x} \qquad (14)\quad -F_{65y}+F_{16y}=m_6a_{G6y}=0$$

The slider's moment equation is not needed: its mass centre is at the pin D and it cannot rotate, so the guide simply supplies whatever couple is required (zero here).

Matrix form $[A]\{B\}=\{C\}$ with the numbers at $\theta_2 = 60°$ ($\mu' = 0$):

Eq.	F12x	F12y	F32x	F32y	F43x	F43y	F53x	F53y	F14x	F14y	F65x	F65y	F16y	T12	{C}
(1)	1	0	1	0	0	0	0	0	0	0	0	0	0	0	−95.538
(2)	0	1	0	1	0	0	0	0	0	0	0	0	0	0	−165.476
(3)	0.0779	−0.0450	−0.0779	0.0450	0	0	0	0	0	0	0	0	0	1	0.000
(4)	0	0	−1	0	1	0	1	0	0	0	0	0	0	0	−650.180
(5)	0	0	0	−1	0	1	0	1	0	0	0	0	0	0	−509.815
(6)	0	0	−0.1108	0.3108	−0.2154	0.2629	0.0446	0.0796	0	0	0	0	0	0	3.462
(7)	0	0	0	0	−1	0	0	0	1	0	0	0	0	0	−278.316
(8)	0	0	0	0	0	−1	0	0	0	1	0	0	0	0	−78.541
(9)	0	0	0	0	0.1786	0.0227	0	0	0.1786	0.0227	0	0	0	0	5.296
(10)	0	0	0	0	0	0	−1	0	0	0	1	0	0	0	−384.577
(11)	0	0	0	0	0	0	0	−1	0	0	0	1	0	0	−149.813
(12)	0	0	0	0	0	0	0.1336	0.3343	0	0	0.1336	0.3343	0	0	1.489
(13)	0	0	0	0	0	0	0	0	0	0	−1	0	0	0	−163.048
(14)	0	0	0	0	0	0	0	0	0	0	0	−1	1	0	0.000

Table 18 – The 14 × 14 coefficient matrix [A] (geometry) and the right-hand side {C} (inertia terms: ma_G in N, $I\alpha$ in N·m).

Solution ($\{B\}=[A]^{-1}\{C\}$, solved with NumPy `linalg.solve`; MATLAB `A\C`, Excel `MMULT(MINVERSE(A),C)` or any matrix solver gives the same):

Joint	Force	x (N)	y (N)	Magnitude (N)	Direction (°)	Magnitude with μ = 0.3 (N)
O (ground–crank)	F_{12}	−1,446.02	−660.07	1,589.55	204.54	1,652.19
A (crank–coupler)	F_{32}	1,350.48	494.60	1,438.20	20.11	1,503.02
B (coupler–rocker)	F_{43}	152.68	49.60	160.53	18.00	167.56

Joint	Force	x (N)	y (N)	Magnitude (N)	Direction (°)	Magnitude with μ = 0.3 (N)
E (ground–rocker)	F_{14}	−125.64	−28.94	128.93	192.97	129.02
C (coupler–rod)	F_{53}	547.62	−64.82	551.45	353.25	627.87
D (rod–slider)	F_{65}	163.05	−214.63	269.54	307.22	339.50
Slider guide	F_{16} (side load)	0.00	−214.63	214.63	270	254.60
Crank	Driving torque T_{12}			**166.005 N·m** (CCW)		177.139 N·m

Table 19 – Joint forces and driving torque at $\theta_2 = 60°$ (forces are those acting on the second-named link, F_ij = force of i on j).

Interpretation. The ground pushes the slider *downwards* ($F_{16}y$ = −214.6 N): the connecting rod pulls the slider up against the top face of the guide. The crank bearing O carries the largest load: $\mathbf{F}_{12} = m_2\mathbf{a}_{G2} - \mathbf{F}_{32}$, i.e. it reacts the 1,438 N force coming from the coupler at A plus the crank's own inertia force (191 N).

Checks.

- *Power (virtual work) balance:* $T_{12}\,\omega_2 = \sum\left(m_i\mathbf{a}_{Gi}\cdot\mathbf{V}_{Gi} + I_i\alpha_i\omega_i\right)$ gives T_{12} = 166.0055 N·m, identical to the matrix solution.
- *Sum of ground reactions:* $\mathbf{F}_{12} + \mathbf{F}_{14} + \mathbf{F}_{16} = \sum m_i\mathbf{a}_{Gi}$ is satisfied to round-off.
- *Independent solver:* the multibody model of Section 7 returns the same pin forces from its Lagrange multipliers (difference < 10^{-7} N).
- *Lecture cross-check:* the Week 9 "Help for 6-bar linkage" slide uses the same OA = 0.18 m and n_2 = 440 rpm. Its link-2 numbers, a_{G2} = 191.08 m/s² at 240°, R_{12} = (−0.045, −0.078) m and R_{32} = (0.045, 0.078) m, are identical to Tables 13 and 17, and its equation (3) is the same as row (3) of Table 18.
- *Sensitivity to the G_3 location:* placing G_3 at the triangle centroid (Table 3 note) instead of on the bisector changes a_{G3} by 1.7 %, T_{12} by 1.2 %, $|F_{12}|$ by 1.5 %, $|F_{32}|$ by 1.7 % and $|F_{43}|$ by 3.4 %. Forces at C, D and the slider are unchanged, because links 5 and 6 do not see link 3's mass distribution.

With Coulomb friction at the slider (μ = 0.3). Friction opposes sliding: $F_{16x} = -\mu\,|F_{16y}|\,\operatorname{sign}(V_D)$. The matrix needs the linear form $F_{16x} = \mu' F_{16y}$, so $\mu' = -\mu\,\operatorname{sign}(V_D)\operatorname{sign}(F_{16y})$, found by solving once with μ′ = 0 to get the signs. Here V_D < 0 and $F_{16}y$ < 0, so μ′ = −0.3 and $F_{16}x$ = +73.16 N (pushing the slider to the right, against its leftward motion). The normal force rises to 243.9 N and the driving torque to 177.14 N·m. The extra 11.13 N·m equals the friction power $|F_{16x}V_D|/\omega_2$. The lecture help example writes $F_{16x} = 0.3F_{16y}$ with a fixed + sign. That is only correct for one combination of slider direction and contact face, so check the sign at every position.

7. Computational Model (Working-Model-equivalent Simulation)

Model. Working Model 2D (WM) is a Windows program that was not available in the environment where this solution was prepared. The rubric accepts "Working Model / MATLAB Simulink / Excel or any other software", so the same model was built from first principles as a planar multibody simulation in Python (`analysis/simulation.py`), exactly the way WM represents it:

- five rigid bodies (links 2–6), each described by the (x, y, φ) of its mass centre → 15 coordinates, with the masses and moments of inertia of Table 3;
- six pin joints (O, A, B, E, C, D), each 2 constraint equations; one keyed-slot joint (slider on y = −H, no rotation), 2 equations; one velocity motor at O ($\varphi_2 = \omega_2 t$), 1 equation → 15 constraints;
- equations of motion $M\ddot{q} + \Phi_q^T \lambda = Q$ with Baumgarte constraint stabilisation, integrated in time with an 8th-order Runge–Kutta solver (DOP853, tolerance 10^{-11}) from a roughly sketched initial configuration that the program "assembles" by Newton–Raphson, as WM does;
- joint forces, the slot reaction (side load) and the motor torque taken from the Lagrange multipliers, the equivalent of WM force meters.

This model does **not** use the closed-form loop equations of Section 5, so it is a genuinely independent check.

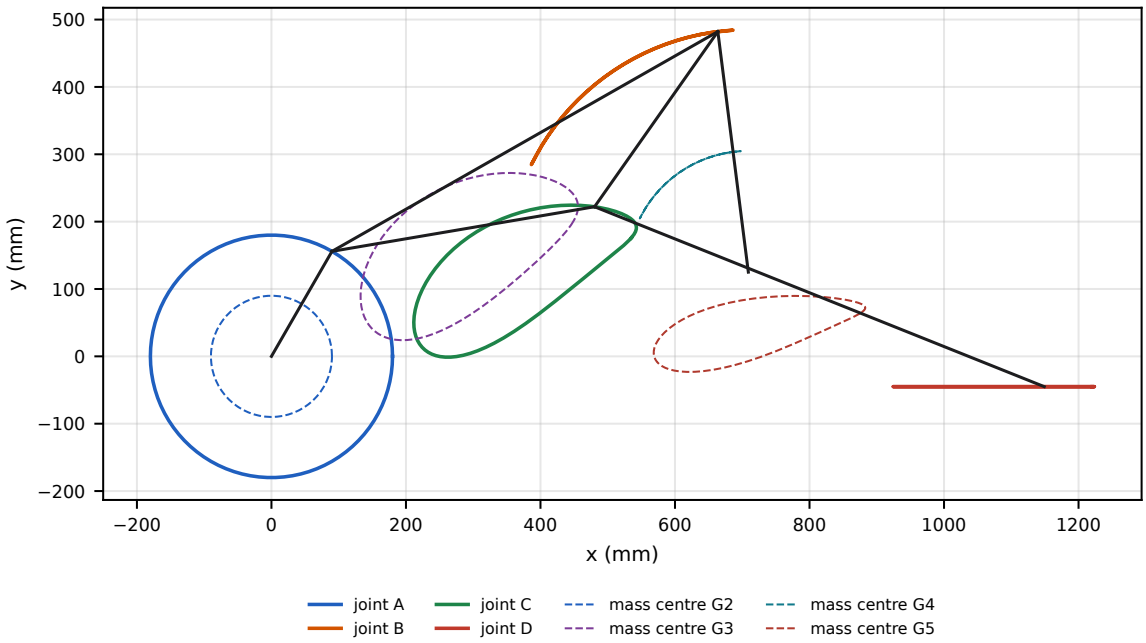


Figure 10 – Computed paths (WM "track" traces) of joints A, B, C, D and mass centres G_2–G_5 over one revolution, with the mechanism at θ_2 = 60°.

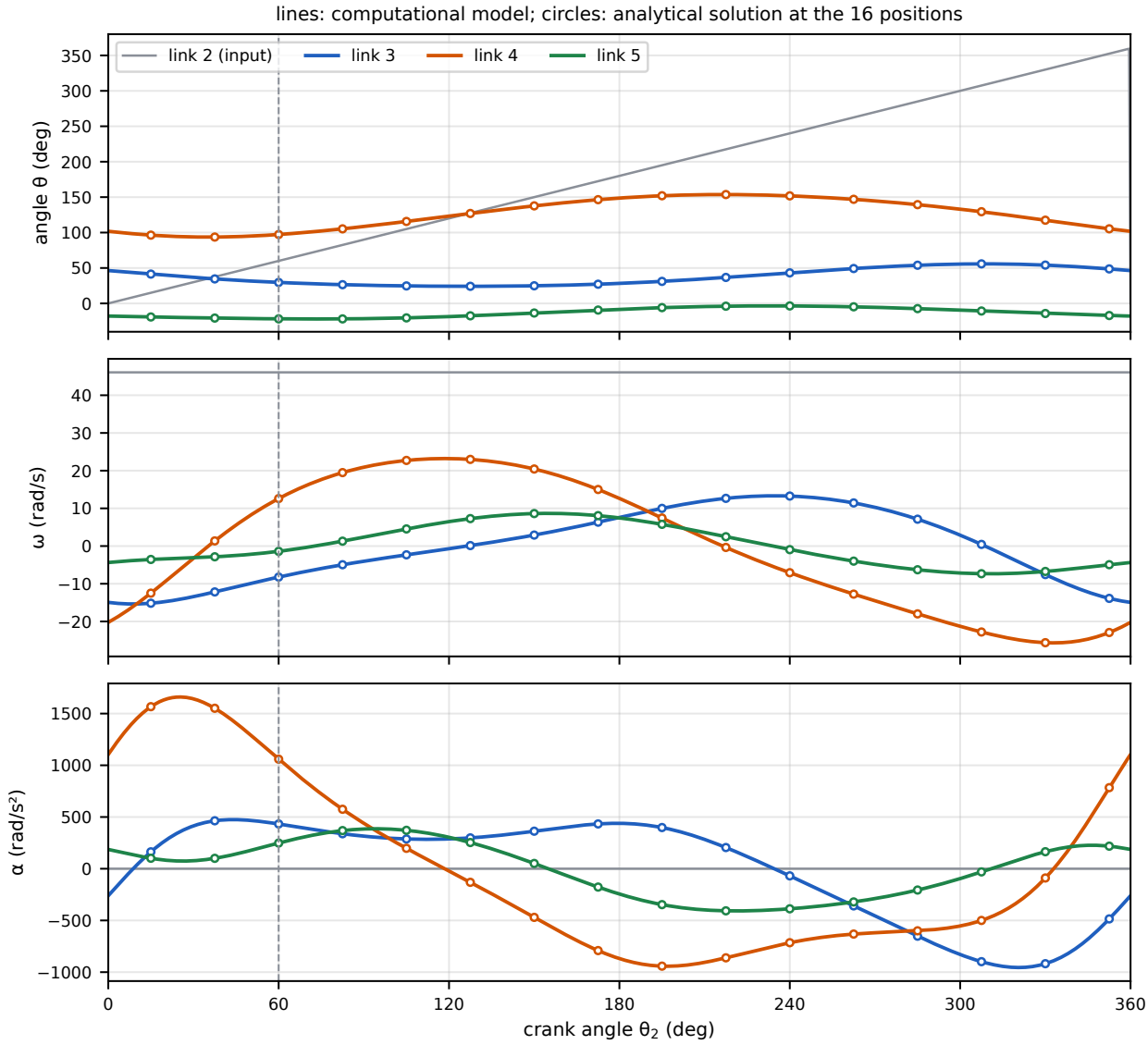


Figure 11 – Angular position, velocity and acceleration of links 3, 4, 5 over one revolution (lines: computational; circles: analytical at the 16 positions; dashed line: $\theta_2 = 60°$).

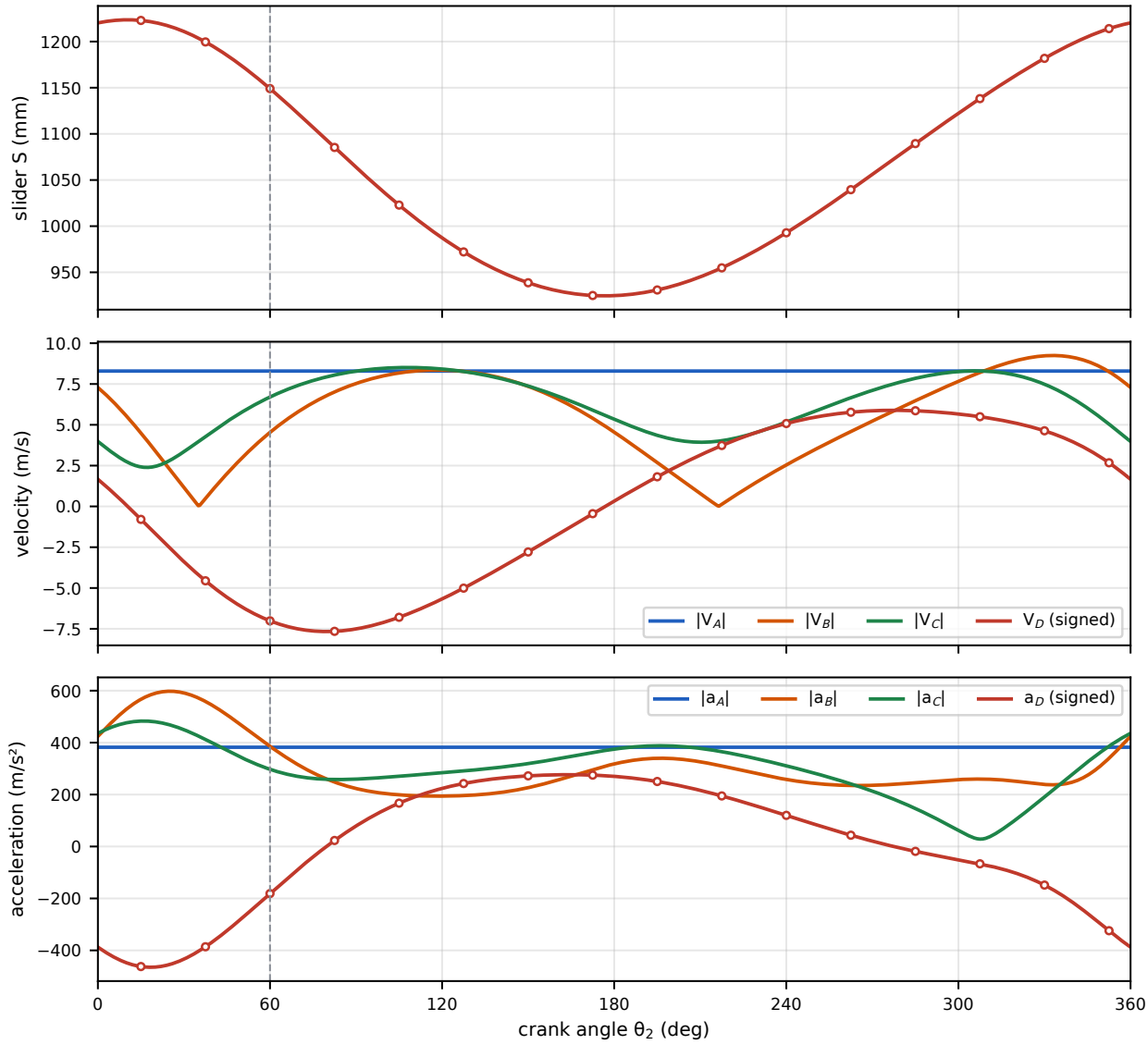


Figure 12 – Slider position, joint velocities and accelerations over one revolution.

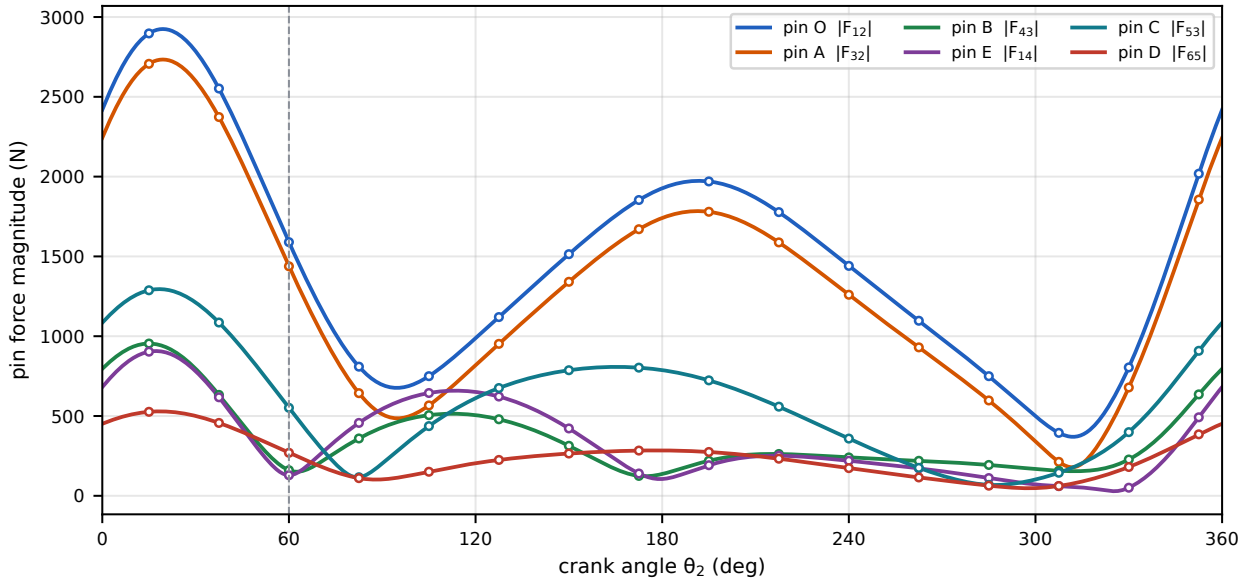


Figure 13 – Pin-force magnitudes at O, A, B, E, C and D over one revolution (μ = 0, no gravity).

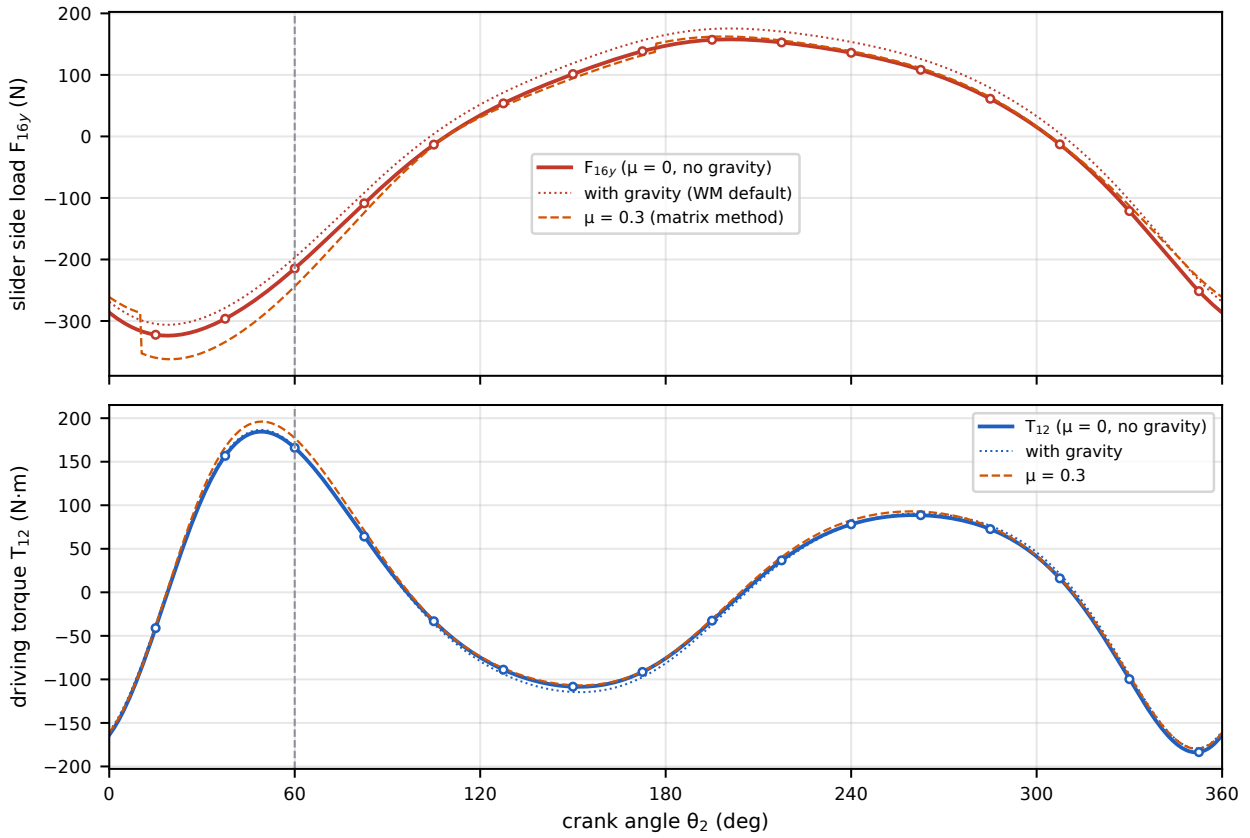


Figure 14 – Slider side load $F_{16}y$ and driving torque T_{12} over one revolution; dotted: with gravity (WM default); dashed: with μ = 0.3 slider friction.

Pos.	θ_2 (°)	θ_3 (°)	θ_4 (°)	θ_5 (°)	S (mm)	ω_3	ω_4	ω_5	V_D (m/s)	α_3	α_4	α_5	a_D (m/s²)
1	60.0	29.627	97.237	−21.776	1,149.04	−8.224	12.599	−1.400	−7.012	432.8	1,059.2	248.2	−181.2
2	82.5	26.444	105.238	−21.844	1,085.30	−4.952	19.484	1.296	−7.656	338.2	574.9	368.8	23.1
3	105.0	24.684	115.672	−20.420	1,022.85	−2.327	22.713	4.543	−6.795	287.6	197.4	371.4	166.9
4	127.5	24.143	126.943	−17.492	972.10	0.125	22.986	7.280	−5.008	298.2	−131.4	253.3	242.5
5	150.0	24.862	137.663	−13.539	938.68	2.913	20.437	8.619	−2.792	362.9	−470.3	50.9	272.0
6	172.5	27.094	146.429	−9.384	924.87	6.335	15.004	8.071	−0.445	433.2	−791.5	−177.4	274.8
7	195.0	31.092	151.961	−5.947	930.86	9.992	7.440	5.765	1.815	398.2	−941.9	−347.0	250.3
8	217.5	36.690	153.660	−3.914	954.84	12.656	−0.366	2.477	3.731	204.8	−861.4	−407.1	194.7
9	240.0	43.112	151.794	−3.548	992.84	13.259	−7.065	−0.949	5.079	−69.0	−715.2	−387.6	119.9
10	262.5	49.243	146.927	−4.779	1,039.53	11.438	−12.749	−4.000	5.770	−359.3	−631.8	−321.1	43.4
11	285.0	53.877	139.412	−7.330	1,089.48	7.123	−17.990	−6.286	5.861	−652.2	−600.3	−206.0	−18.8
12	307.5	55.809	129.420	−10.717	1,138.16	0.439	−22.798	−7.336	5.496	−898.7	−500.1	−30.7	−66.7
13	330.0	54.079	117.454	−14.222	1,181.81	−7.562	−25.634	−6.737	4.633	−918.2	−89.6	164.4	−148.6
14	352.5	48.693	105.290	−17.093	1,214.03	−13.877	−22.930	−4.944	2.675	−485.2	783.7	218.1	−324.2
15	15.0	41.372	96.364	−19.125	1,222.90	−15.168	−12.494	−3.554	−0.796	163.3	1,566.8	101.3	−462.2
16	37.5	34.594	93.655	−20.687	1,199.60	−12.168	1.362	−2.846	−4.559	463.6	1,551.4	100.5	−386.1

Table 20 – Computational kinematics at the 16 positions (ω in rad/s, α in rad/s², + = CCW; identical to Table 15 to all printed digits).

Pos.	θ_2 (°)	T_{12} (N·m)	F_{16}y side load (N)	pin O F_{12} (N)	pin A F_{32} (N)	pin B F_{43} (N)	pin E F_{14} (N)	pin C F_{53} (N)	pin D F_{65} (N)
1	60.0	166.01	−214.6	1,589.5	1,438.2	160.5	128.9	551.4	269.5
2	82.5	64.08	−108.7	809.3	643.2	359.3	456.8	116.3	110.7

Pos.	θ_2 (°)	T_{12} (N·m)	F_{16}y side load (N)	pin O F_{12} (N)	pin A F_{32} (N)	pin B F_{43} (N)	pin E F_{14} (N)	pin C F_{53} (N)	pin D F_{65} (N)
3	105.0	−33.22	−13.3	749.6	566.4	505.3	644.2	436.7	150.8
4	127.5	−88.88	53.7	1,119.9	952.1	479.6	622.0	675.3	224.8
5	150.0	−108.42	101.4	1,514.1	1,340.9	313.4	422.1	786.9	265.0
6	172.5	−91.44	138.5	1,853.4	1,670.4	124.7	139.7	803.0	283.5
7	195.0	−32.54	157.1	1,969.9	1,779.7	219.6	191.7	723.4	274.6
8	217.5	36.69	152.8	1,777.5	1,587.8	262.0	251.2	558.7	232.5
9	240.0	78.14	136.0	1,440.7	1,259.8	240.3	219.7	358.6	173.6
10	262.5	88.58	108.2	1,096.9	930.1	219.3	172.1	175.1	115.1
11	285.0	72.63	61.2	749.7	597.5	193.4	111.2	68.4	63.5
12	307.5	16.04	−12.8	394.5	212.8	157.0	60.6	144.4	61.4
13	330.0	−99.73	−121.3	804.4	678.8	227.4	50.8	398.9	180.5
14	352.5	−183.56	−251.5	2,018.3	1,855.9	635.5	492.0	908.6	385.2
15	15.0	−41.09	−322.4	2,897.2	2,706.8	954.0	903.1	1,287.9	526.3
16	37.5	156.73	−296.3	2,552.0	2,373.3	632.3	617.4	1,085.9	456.7

Table 20b – Computational driving torque, slider side load and pin-force magnitudes at the 16 positions (μ = 0, no gravity).

Extreme values over one revolution.

Quantity	Value	at θ_2
Driving torque, maximum / minimum	+184.6 / −183.8 N·m	49.0° / 351.5°
Driving torque, mean / RMS	0.000 / 97.7 N·m	–
Slider side load, largest magnitude	323.8 N	19.0°
Pin O (crank bearing) force, maximum	2,925 N	19.5°
Pin A force, maximum	2,734 N	19.5°
Pin B force, maximum	954 N	15.0°
Pin E (rocker bearing) force, maximum	907 N	17.0°
Pin C force, maximum	1,295 N	18.5°
Pin D force, maximum	529 N	18.5°

Table 21 – Peak loads over one revolution (computational model).

How do the results compare with the analytical and graphical approaches?

Quantity	max |computational − analytical| over the revolution	Largest value
theta3 (deg)	4.4e-10	55.815
theta4 (deg)	6.4e-10	153.664
theta5 (deg)	4.0e-10	21.979
S (m)	2.1e-12	1.224
omega3 (rad/s)	1.7e-10	15.367
omega4 (rad/s)	4.5e-10	25.684
omega5 (rad/s)	1.5e-10	8.667
V_D (m/s)	7.0e-11	7.668

Quantity	max |computational − analytical| over the revolution	Largest value
alpha3 (rad/s2)	8.9e-09	955.387
alpha4 (rad/s2)	1.0e-08	1,660.368
alpha5 (rad/s2)	5.1e-09	408.051
a_D (m/s2)	3.7e-09	464.991
F12 magnitude (N)	7.8e-09	2,924.718
F32 magnitude (N)	8.3e-09	2,733.659
F43 magnitude (N)	1.2e-08	954.026
F14 magnitude (N)	1.4e-08	907.126
F53 magnitude (N)	8.5e-09	1,295.087
F65 magnitude (N)	3.4e-09	529.141
F16y (N)	2.3e-09	323.822
T12 (N m)	2.5e-09	184.581

Table 22 – Agreement between the independent computational model and the analytical solution at all 721 time steps (0.5° of crank rotation).

- **Computational vs analytical:** identical to better than 10^{-7} in every kinematic and dynamic quantity (Table 22). Two completely different formulations agreeing this closely confirms both. The mean driving torque over a cycle is 0.0000 N·m, i.e. zero. This is correct physics: with no load and no friction, the energy the motor puts in during part of the cycle is returned during the rest.
- **Computational vs graphical:** within the drawing accuracy (Table 16), ≤ 0.8 % for velocities and ≤ 0.2 % for accelerations.
- **Gravity.** Working Model has gravity switched on by default. Including it changes T_{12} by at most 6.24 N·m and the side load by at most 17.7 N (at 60°: T_{12} = 167.38 N·m, $F_{16}y$ = −197.0 N). Switch gravity **off** (World ▸ Gravity ▸ None) when comparing with the matrix method.
- **Friction.** With μ = 0.3 the torque peak rises to 196.1 N·m and the mean torque becomes 3.66 N·m, the friction loss. The side load jumps when the slider reverses (θ_2 = 10.4° and 176.8°), because the friction force changes direction.
- **Expected WM differences.** A real WM run typically differs from the analytical values by 0.1–1 %. The reasons are its integrator and animation step, meter rounding, gravity being left on, or measuring at a body's mass centre instead of at the joint. The FAQ notes that simulated results about 50 % lower than the analytical ones are the typical symptom of measuring at the COM. Put point elements on the joints before measuring.

7.1 How to build this model in Working Model 2D

1. **Set up:** View ▸ Workspace: tick Grid Lines and X,Y Axes. View ▸ Numbers and Units: SI (degrees); More Choices ▸ Rot. Velocity = rev/min (to show rpm). World ▸ Gravity: None (for comparison with Section 6).
2. **Draw the bodies at θ_2 = 60°** using the coordinates below (origin at O). Draw links 2, 4 and 5 as thin rectangles (e.g. 10 mm wide) between their joint points, link 3 as a polygon A-B-C, and the slider as a small rectangle at D. The ground parts are anchored (anchor tool): pivot O, pivot E and the slider track at y = −H.
3. **Mass properties (Window ▸ Properties):** enter the masses and moments of inertia of Table 3. Link 3's centre of mass must be at G_3, not at the polygon centroid: in Window ▸ Geometry set the COM offset. At the drawn position, G_3 − centroid = (−10.55, −19.99) mm in global axes (convert to the body's own axes if the polygon was drawn rotated).
4. **Joints:** pin joints at O (link 2–ground), A (2–3), B (3–4), E (4–ground), C (3–5) and D (5–6); a keyed slot joint between the slider and the horizontal track. Select all links ▸ Object ▸ Do Not Collide (FAQ).

5. **Motor:** at O, type Velocity, 440 rpm (positive = anticlockwise).
6. **Frames:** World ▸ Accuracy ▸ Animation step = T/16 = 60/(440 × 16) = 0.0085227 s. The FAQ notes that WM rounds this box to 3 decimals: here 0.009 s, which is 23.76° per frame instead of 22.5°, so 16 frames overshoot one revolution by 20°. Treat the frames as a quick visual check only, and take the exact 16 positions from a crank-angle meter, or from Tables 4 and 20. World ▸ Pause Control: `frame() = 16`. Without rounding, frame k (k = 0 … 15) of a model drawn at 60° would be position k + 1. For the FAQ's six-frame method (model drawn at $\theta_2 = 0°$), use T/6 = 0.022727 s; WM rounds it to 0.023 s, so the first frame lands at 60.7° rather than exactly 60°.
7. **Meters:** put Point elements (snapped, "X") on A, B, C, D and on each mass centre. Then use Measure ▸ Position / Velocity / Acceleration on the points, Measure ▸ Force on the pins and slot, and Measure ▸ Torque on the motor. Joint paths: select a point ▸ Appearance ▸ Track centre of mass.

Point	x at $\theta_2 = 60°$ (mm)	y at $\theta_2 = 60°$ (mm)	x at $\theta_2 = 0°$ (mm)	y at $\theta_2 = 0°$ (mm)
O	0.00	0.00	0.00	0.00
A	90.00	155.88	180.00	0.00
B	663.71	482.16	635.66	477.46
C	480.42	222.11	534.89	175.69
D	1,149.04	−45.00	1,220.23	−45.00
E	709.06	125.03	709.06	125.03
G2	45.00	77.94	90.00	0.00
G3	400.83	266.73	445.83	195.54
G4	686.39	303.59	672.36	301.25
G5	814.73	88.56	877.56	65.35
angles θ_3, θ_4, θ_5	29.627°, 97.237°, −21.776°		46.338°, 101.764°, −17.850°	

Table 23 – Working Model build coordinates (origin at O, x right, y up).

8. Shaking Force, Shaking Moment and Balancing

8.1 Shaking force and moment at $\theta_2 = 60°$ (course template)

The forces of the moving links on the frame are $\mathbf{F}_{21} = -\mathbf{F}_{12}$, $\mathbf{F}_{41} = -\mathbf{F}_{14}$ and $\mathbf{F}_{61} = -\mathbf{F}_{16}$ (Table 19):

	x (N)	y (N)
F_{21} (at O)	1,446.02	660.07
F_{41} (at E)	125.64	28.94
F_{61} (at D, along the guide)	0.00	214.63
Sum = shaking force F_s	**1,571.66**	**903.65**

Table 24 – Forces on the frame at $\theta_2 = 60°$.

$$F_s = \sqrt{\sum F_x^2 + \sum F_y^2} = \sqrt{1,571.7^2 + 903.6^2} = 1,812.9 \text{ N},$$

$$\phi = \tan^{-1}\frac{903.6}{1,571.7} = 29.90°.$$

The shaking (reaction) torque is $T_s = T_{21} = -T_{12} = -166.01$ N·m, and the shaking moment about O is

$$M_s = T_{21} + \mathbf{R}_E \times \mathbf{F}_{41} + \mathbf{R}_D \times \mathbf{F}_{61} = -166.01 + 4.81 + 246.62 = 85.43 \text{ N}\cdot\text{m}\ (\text{CCW}).$$

Check: $\mathbf{F}_s = -\sum m_i \mathbf{a}_{Gi}$ and $M_s = -\sum(\mathbf{r}_{Gi} \times m_i\mathbf{a}_{Gi} + I_i\alpha_i)$ give the same values. Friction at the slider does not change $\mathbf{F}_s$ or M_s, only how the load is shared between the bearings.

8.2 Over one revolution

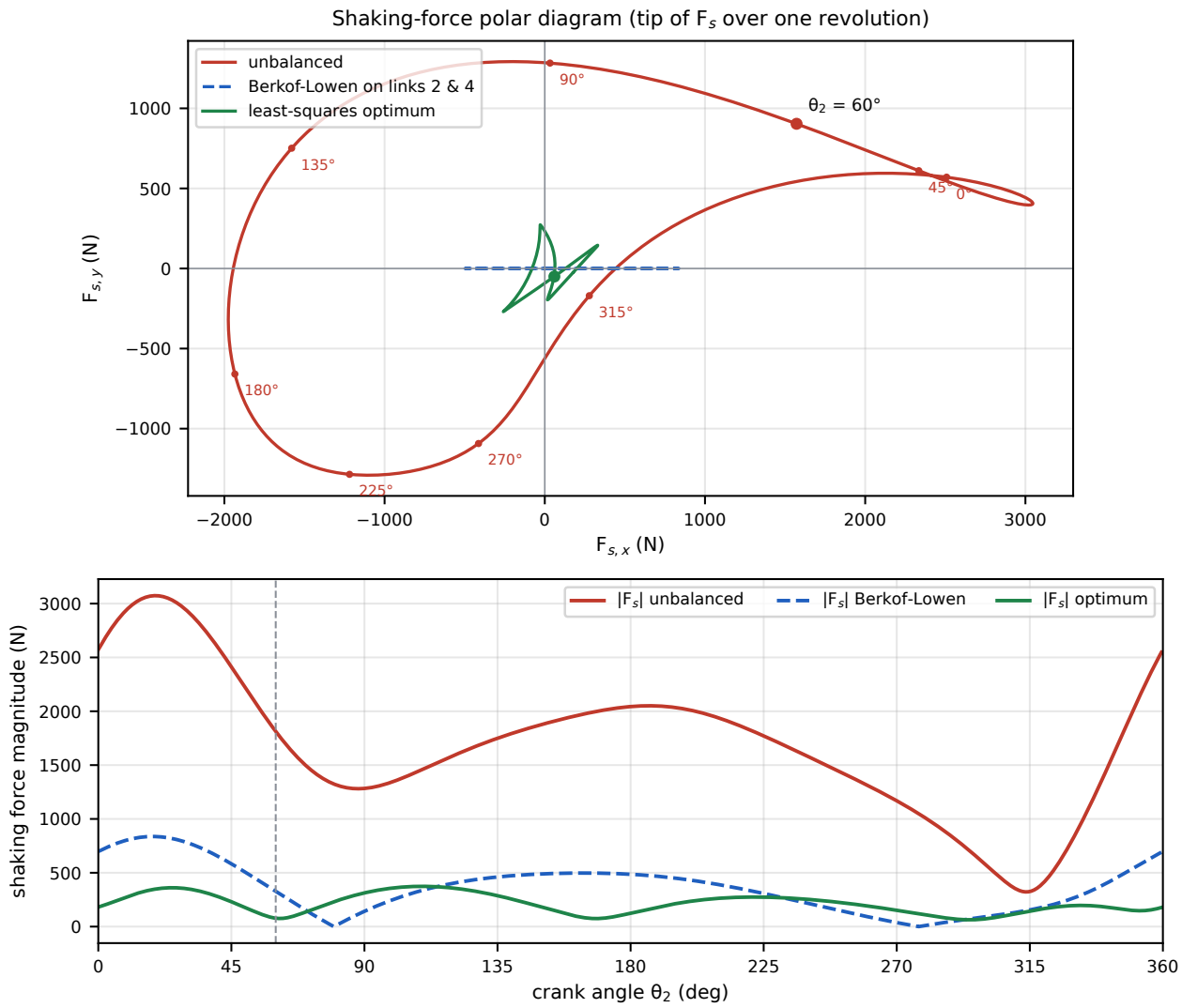


Figure 15 – Shaking force: polar diagram of F_s (top; crank angles marked on the unbalanced curve) and its magnitude over one revolution (bottom), unbalanced and after balancing.

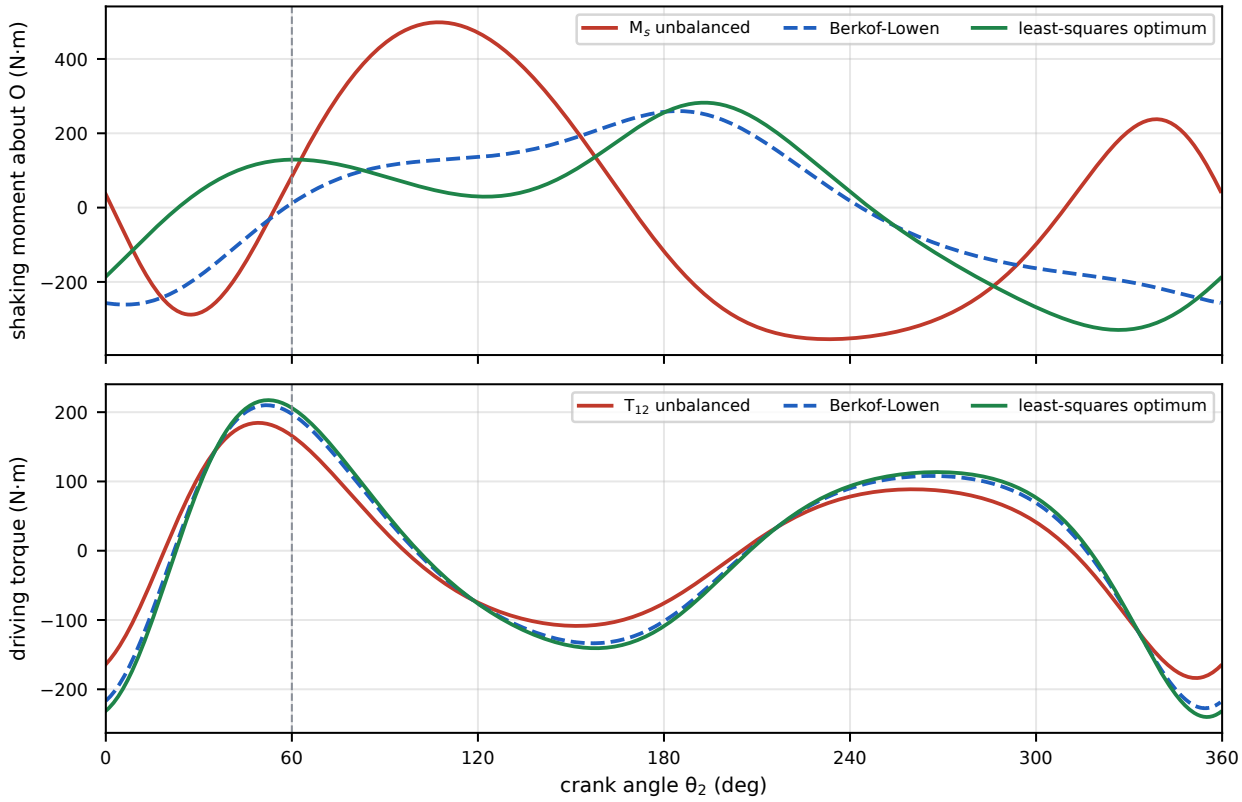


Figure 16 – Shaking moment about O and driving torque over one revolution, before and after balancing.

Unbalanced, the shaking force reaches 3,074 N (RMS 1,803 N). It is dominated by its horizontal component (up to 3,047 N), which comes from the large horizontal accelerations of links 3, 5 and the slider. The shaking moment reaches 499 N·m.

8.3 Analytical balancing strategy

A mechanism exerts no shaking force when the **total centre of mass of its moving links is stationary**, because $\mathbf{F}_s = -M_{tot}\,\mathbf{a}_{cm}$. The strategy below makes this as nearly true as possible using counterweights on the two links that are pivoted to the frame (crank 2 and rocker 4), since they are the only places where added mass can be supported by an existing bearing.

Step 1 – Model link 5 by two point masses. Replace link 5 by $m_5/2 = 0.90$ kg at C and $m_5/2$ at D. This has the same mass and centre of mass, so it gives exactly the same shaking force. Link 3 then carries $m_3' = m_3 + m_5/2 = 3.40$ kg with its mass centre, in a frame fixed to link 3 (origin A, x along AB), at

$$g_3 = \frac{m_3\,\frac{l_3}{2}e^{-j\beta/2} + \frac{m_5}{2}\,\overline{AC}\,e^{-j\beta}}{m_3'} = 0.3375 - j\,0.0780\ \text{m}.$$

The reciprocating mass at D is $m_{rec} = m_6 + m_5/2 = 1.80$ kg.

Step 2 – Berkof–Lowen balance of the four-bar O-A-B-E. Write the total mass moment of links 2, 3′ and 4 using body-fixed complex vectors (P_2 for link 2 about O, P_4 for link 4 about E):

$$M\mathbf{r}_{cm} = P_2 e^{j\theta_2} + m_3'\left(l_2 e^{j\theta_2} + g_3 e^{j\theta_3}\right) + m_4\mathbf{R}_E + P_4 e^{j\theta_4}.$$

Eliminate $e^{j\theta_3}$ with loop 1, $e^{j\theta_3} = (l_1 e^{j\theta_1} + l_4 e^{j\theta_4} - l_2 e^{j\theta_2})/l_3$:

$$M\mathbf{r}_{cm} = \left[P_2 + m_3' l_2\left(1 - \tfrac{g_3}{l_3}\right)\right]e^{j\theta_2} + \left[P_4 + m_3'\tfrac{g_3}{l_3}l_4\right]e^{j\theta_4} + \text{constant}.$$

The centre of mass is stationary for every θ_2 if both brackets vanish:

$$P_{2,req} = -m_3' l_2 \left(1 - \frac{g_3}{l_3}\right) = -0.2991 - j\,0.0723 \ \text{kg}\cdot\text{m},$$

$$P_{4,req} = -m_3' \frac{g_3\, l_4}{l_3} = -0.6258 + j\,0.1446 \ \text{kg}\cdot\text{m}.$$

Subtracting what the links already have ($m_2 l_2/2 = 0.090$ kg·m and $m_4 l_4/2 = 0.270$ kg·m along the links) gives the counterweight mass moments

$$m_{c2}\,\mathbf{u}_2 = -0.3891 - j\,0.0723 \ \text{kg}\cdot\text{m} \ (|\,| = 0.396),$$

$$m_{c4}\,\mathbf{u}_4 = -0.8958 + j\,0.1446 \ \text{kg}\cdot\text{m} \ (|\,| = 0.907).$$

In practice: a 3.30 kg mass at 120 mm on the crank, almost opposite A (angle 190.5° from OA), and a 7.56 kg mass at 120 mm on an extension of the rocker beyond E (angle 170.8° from EB). A larger radius reduces the masses in proportion (m·r is what matters).

Step 3 – The residual. After Step 2, $\mathbf{F}_s = -m_{rec}\,a_D$, which is purely horizontal (verified numerically to 10^{-12}). It cannot be cancelled by a rotating counterweight, which produces a force of constant magnitude rotating at crank speed. Such a counterweight can only remove the forward-rotating half of the first harmonic of a_D, and it trades horizontal for vertical force. The best amount is found by least squares (Step 4).

Step 4 – Optimum. Because the counterweights do not change the kinematics, their inertia forces are *linear* in the complex mass moments: $\mathbf{F}_s(\theta) = \mathbf{F}_{s0}(\theta) + P_2\,\omega_2^2 e^{j\theta_2} - P_4\,(j\alpha_4 - \omega_4^2)e^{j\theta_4}$. Minimising $\sum_\theta |\mathbf{F}_s|^2$ over one revolution is therefore a linear least-squares problem for P_2 and P_4, with optimum $P_2 = -0.4748 - j\,0.1239$ and $P_4 = -1.0974 + j\,0.3591$ kg·m.

Strategy	Counterweights (at 120 mm)	RMS F_s (N)	Peak F_s (N)	Peak M_s (N·m)	Torque range (N·m)
Unbalanced (as designed)	–	1,803	3,074	499	−184 … +185
Crank counterweight only (least squares)	crank 6.19 kg, rocker 0.00 kg	874	1,619	499	−184 … +185
Berkof–Lowen on links 2 & 4 (exact 4-bar balance)	crank 3.30 kg, rocker 7.56 kg	422	837	261	−227 … +210
Berkof–Lowen + extra crank weight (least squares)	crank 4.38 kg, rocker 7.56 kg	315	574	261	−227 … +210
Global least-squares optimum on links 2 & 4	crank 4.09 kg, rocker 9.62 kg	231	374	329	−240 … +217

Table 25 – Comparison of balancing strategies over one revolution.

8.4 Discussion

- The **Berkof–Lowen** counterweights remove all of the shaking force except the slider's reciprocating part. The RMS falls by 77 % and the peak by 73 %, and the remaining force is purely horizontal, which is easier to absorb in the machine frame. The shaking moment also falls, by 48 % in peak.
- The **least-squares optimum** cuts the RMS shaking force by 87 %. It does so by over-balancing the crank and rocker slightly, so that part of the reciprocating force is cancelled at the cost of a small vertical component.
- **Costs.** About 13.7 kg of counterweights is added to a mechanism whose moving links weigh 7.7 kg. The driving-torque range grows from ±185 to −240 … +217 N·m, and the bearings at O and E carry the counterweight inertia.

A flywheel on the crank shaft is recommended to keep the speed constant, and the bearings should be sized for the larger loads.

- **Complete balancing** would also require cancelling the reciprocating force. One option is a mirror-image second slider driven 180° out of phase from the same crank (an opposed arrangement). Another is a dedicated balancing slider. A counter-rotating inertia geared to the crank can remove the shaking moment. These add complexity, and are worthwhile only if the residual of about 374 N peak is still too large for the application.
- **Recommended strategy:** fit Berkof–Lowen counterweights on links 2 and 4, then trim the crank counterweight towards the least-squares value, plus a flywheel. This removes most of the shaking force with two simple parts and keeps the remaining force horizontal.

Conclusions

1. The selected dimensions (AB = 660 mm, BE = 360 mm) satisfy the course selection rules and Grashof's criterion. The mechanism is a crank–rocker driving a slider through a 299.1 mm stroke with a quick-return ratio of 1.16, and the transmission angle never falls below 54.8°.
2. At $\theta_2 = 60°$ the slider moves left at 7.012 m/s and is still speeding up (a_D = 181.2 m/s² to the left), and the links' angular velocities are $\omega_3 = 8.22$ rad/s CW, $\omega_4 = 12.60$ rad/s CCW and $\omega_5 = 1.40$ rad/s CW. The angular accelerations are $\alpha_3 = 432.8$, $\alpha_4 = 1{,}059.2$ and $\alpha_5 = 248.2$ rad/s², all CCW.
3. The four independent methods (graphical polygons, instantaneous centres, analytical vector loops and the computational model) agree, within 0.8 % (velocity) and 0.2 % (acceleration) for drawings read at ruler precision, and to 10^{-7} between the analytical and computational models.
4. At $\theta_2 = 60°$ the matrix method gives a driving torque of 166.0 N·m and a largest pin force of 1,590 N at the crank bearing. Over a revolution the torque swings ±185 N·m and the crank bearing peaks at 2,925 N.
5. The unbalanced shaking force (up to 3,074 N) can be reduced by 87 % (RMS) with two counterweights. The residual reciprocating force of the slider can only be removed with an opposed or balancing slider.

References

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Appendix A – Notes on the lecture examples

While checking this solution against the lecture material, three slips were found in the worked examples. They are listed here so that they do not cause confusion; the *methods and formulas* in the notes are correct.

1. **Week 7, four-bar position example** (a = 40, b = 120, c = 80, d = 100 mm, θ_2 = 40°). The handwritten result gives θ_4(open) = 98° and θ_4(crossed) = −57°. Substituting back into the loop shows |AB| = 79 mm for θ_4 = 98°, not 120 mm. The slip is the extra minus sign written in front of 2 $\tan^{-1}$(…). Without it, the two roots of the same quadratic are θ_4 = −98.0° and θ_4 = **+57.3°**, and the root that pairs with θ_3 = 20.2° (the configuration used in the velocity and acceleration examples) is θ_4 = 57.3°; the other pair is θ_3 = −60.9°, θ_4 = −98.0°.
2. **Week 7, velocity example** (ω_2 = 25 rad/s). ω_4 is evaluated with sin(40° − 23.3°) instead of sin(40° − 20.3°). The correct value is ω_4 = **7.00 rad/s**, not 5.98 rad/s. The acceleration example inherits this (and uses 45 instead of a = 40 in F), so it should read α_3 ≈ **296 rad/s²** and α_4 ≈ **470 rad/s²**. Both were confirmed by numerical differentiation.
3. **Week 7, template Part 3 (slider).** The template writes $Y_C + l_5 \sin\theta_5 = H$ but also $Y_D = -H$ for a slider below O. For the handout geometry, use $Y_C + l_5 \sin\theta_5 = -H$, i.e. $\theta_5 = \arcsin[-(H + Y_C)/l_5]$ (Section 5.1).

Always check a computed configuration by substituting it back into the loop equations.

Appendix B – Software and reproducibility

All calculations, drawings and this report are generated by the Python scripts in the repository:

File	Purpose
`analysis/params.py`	the four given parameters, the selected AB, BE and the data-selection equations
`analysis/kinematics.py`	analytical (complex-number) positions, velocities and accelerations (Section 5)
`analysis/sec1_position.py` … `sec4_ic.py`	graphical constructions drawn to scale (Sections 1–4)
`analysis/dynamics.py`, `sec6_fbd.py`	14 × 14 matrix method, FBDs, shaking force and moment (Section 6)
`analysis/simulation.py`, `sec7_simulation.py`	independent multibody (Working-Model-equivalent) simulation (Section 7)
`analysis/sec8_balancing.py`	shaking force/moment over a revolution and balancing optimisation (Section 8)
`analysis/run_all.py`, `report/build_report.py`	run everything and rebuild this report

To use your own parameters, edit the four given values in `analysis/params.py` (and choose AB and BE that satisfy the inequalities), then run `python3 analysis/run_all.py` and `python3 report/build_report.py`.